MARPAC RDIMS # 815141

ROYAL CANADIAN NAVY
QUALIFICATION STANDARD
AND PLAN (QSP)

Marine Technician (Mar Tech) (00379)

USQ-PO2-MAR TECH Mechanical Supervisor
MITE Course ID: 122003

USQ-PO2-MAR TECH Mechanical Supervisor - ALUX

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Fleet School Pacific (NFS(P)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0 (QSPWB Apr 2018)	10 May 2024	CNTG

TABLE OF CONTENTS

RECORD OF AMENDMENTS 1
TABLE OF CONTENTS 2
FOREWORD 4
CHAPTER 1 TRAINING MANAGEMENT DETAILS 1-1
AIM 1-2
COURSE DURATION 1-2
RESOURCES 1-2
COURSE CAPACITY 1-3
MITE 1-3
COURSE REPORTING 1-4
PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR) 1-4
SPECIAL INSTRUCTIONS 1-4
CHAPTER 2 ASSESSMENT OF TRAINEE 2-1
GENERAL 2-2
ENABLING AND PERFORMANCE CHECKS 2-2
RESPONSIBILITY FOR TESTING 2-3
PROGRESS MONITORING 2-3
TRAINING REVIEW BOARD 2-3
COURSE REPORTS 2-4
CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS ... 3-1
PO 001 Assess Ship's Stability 3-2
EO 001.01 Identify stability states and forces acting on a ship 3-4
EO 001.02 Extract Information from hydrostatic curves 3-8
EO 001.03 Calculate mean draught of a ship following a change of weight 3-10
EC 001.03: Calculate mean draught of a ship following a change of weight 3-14
EO 001.04 Calculate the ship's centre of gravity 3-15
EC 001.04: Calculate the ship's centre of gravity 3-19
EO 001.05 Calculate the free surface effect in a ship's compartment 3-20
EO 001.06 Calculate the rise in gravity due to FCE 3-22
EC 001.06: Calculate rise in G due to FSE and FCE 3-25
EO 001.07 Calculate stability following an abnormal condition 3-26
EC 001.07: Calculate a ship's stability due to an abnormal condition 3-30
PC 001: Assess ship's stability 3-31

MARPAC RDIMS #815141
Chap, 1, approved 10 05 2024
Chap 2, 3approved 10 05 2024

PO 002 Perform the Duties of Fuelling I/C. 3-32
EO 002.01 Initiate fuelling orders 3-34
EC 002.01: Complete Fuelling Orders 3-38
EO 002.02 Lead Maritime Vessel Fuelling 3-39
EC 002.02: Lead Vessel Fuelling 3-43
PC 002: Lead Maritime Vessel Fuelling 3-46
PO 003 Manage Battle Damage Control Organization IT0350 3-49
EO 003.01 Perform DCPO duties 3-50
EC 003.01: Perform DCPO duties 3-55
EO 003.02 Submit a Post-Incident Report 3-1
EC 003.02: Submit Post-incident Report 3-3
PC 003: Manage Battle Control Organization 3-6
ANNEX A - MAIN REFERENCES A-1
ANNEX B - TASK LIST B-1
ANNEX C - QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS C-1
ANNEX D - QSPWB SCALAR D-1
ANNEX E - ON JOB PERFORMANCE RECORD (OJPR) E-1
ANNEX F - ACRONYMS AND ABBREVIATIONS F-1
ANNEX G - SUGGESTED SEQUENCING AND TIMINGS G-1
ANNEX H - ADDITIONAL TRAINING RESOURCES H-1
ANNEX I - COSTING INFORMATION I-1
ANNEX J - COURSE HISTORY J-1
ANNEX K - ASSESSMENT PLAN K-1
ANNEX L - TEL PLAN L-1

FOREWORD

1. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification 9 Feb 11;

c. A-PD-055-002/PP-002Marine Technician MOSID 00379 Occupational Specification 15 Dec 16;

d. NAVORD 4500-0 “Structured Delegation”; and

e. NPTORD 4500-22 “Qualification Standard and Plan Production and Management”.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the NFS(P) (MS Div), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training are indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	122003	ALUX
NFS(A)	All	122003	ALUX

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Marine Technician Mechanical Supervisor. This QSP is designed to assist the training system in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

Allocation of ATT	**Minutes**
(Sum of all EOs/POs)	7715
Enabling Checks (ECs)	2210
Performance Checks (PCs)	1410
Actual Training Time (ATT) Sub-total	**11335**
Allocation of BAT, TAT, TTT, PT(Course >31 days)	
Base Administrative Time (BAT)	120
Training Administrative Time (TAT)	120
Training Travel Time (TTT)	0
Physical Fitness (PT)	1020.15
Sub-total	**1260.15**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	**12595.15**
Total mins/400 = # days (rounded up)	
Training Days	**32**

RESOURCES

1. The following table details key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Sr Instructor	CPO2/PO1	Dept Coord or Mar Tech Mgr	1	All
Stability Instructor	Lt(N) - PO1 or civilian	• Lt(N) MS Eng AHOD Qual • CPO1 – PO1 legacy HT or Mar Tech Mech Mgr • Civilian – academic instructor with stability background	1	PO 001

Fuelling Instructor	PO1/PO2	• Legacy PO1/PO2 Mar Eng with Cert 3 • Mar Tech PO1/PO2 Mechanical Mgr/Supr	1	PO 002
DCPO Instructor	CPO2/PO1	Mar Tech RQ-PO1 (Mechanical) with tour as DCPO	2	PO 003
Nav Arch Guest Speaker	Lt(N) Or Civilian	• Lt(N) MS Eng with Nav Arch qual • Civilian with Nav Arch qual	1	PO 001

2. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP.

Critical Training Resources	PO/EO
DWAN Laptop (each trainee)	All
DWAN drops	All
Scientific Calculator (each trainee)	PO 001
MT&S (each trainee)	PO 001/003
BDCS trainer	PO 002/003
PCOT trainer	PO 002
SHINCOM	PO 003
DCTFPO	PO 003

COURSE CAPACITY

3. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C.

a. Maximum – 10;

b. Minimum – 6; and

c. Optimum – 10.

MITE

4. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

5. Scheduling. Each session of this course shall be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training

breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

6. Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

a. Mar Tech RQ PO2; and

b. Minimum rank PO2.

7. Qualification. Successful completion of this course constitutes the primary means of attaining the Marine Technician Mechanical Supervisor (National Qualification(s) MITE). Additionally, the following MITE Course IDs will be granted: nil.

COURSE REPORTING

8. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

9. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

SPECIAL INSTRUCTIONS

10. Lesson Objectives throughout this QSP are rated from Low to High priority for development in the standards paragraphs of each EO and PO. Criteria for rating is IAW the following:

a. Low: Impact to the trainees is low if this training is not completely developed. These LOs are relatively easy and frequently performed or theory-based where proficiency can be gleaned from the TPt table, references, and/or experience;

b. Medium: LOs require further development than present in the TPt table /references. Instruction must be to a level where the trainees can accomplish the task competently; and

c. High: Impact to the trainees is high if this training is not effectively developed. These LOs are generally related to safety where retention is critical to minimize the likelihood of skill fade.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL FLEET SCHOOL PACIFIC

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

3. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

4. Enabling Checks (ECs). ECs are formative assessments that help the trainee, and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

5. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol 6 with the unsatisfactory progress documented in their Trainee File.

6. Critical Enabling Checks (CECs). CECs are summative assessments at the end of an EO which are considered critical because trainees cannot progress in the training without first successfully completing them. They are related to elements such as safety and security where harm could result from poor performance. CECs are treated the same as PCs.

7. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

8. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

9. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

10. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

11. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NFS(P).

PROGRESS MONITORING

12. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

13. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

14. Continuous monitoring of candidate progress is required to provide the following:

    a. Early warning of trainee difficulties;

    b. Feedback on the effectiveness of training;

    c. Information for any required TRBs; and

    d. Information to assist in completion of the CF 377 Course Report.

15. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

16. Instructional staff shall maintain individual trainee files, including the results of all:

    a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

    b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

    c. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

17. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

18. In the event of prolonged trainee absence, TE Staff shall determine, on a case-by-case basis, the necessity to convene a TRB.

COURSE REPORTS

19. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51“Course Reports and Certifications”.

20. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

21. The Course Report narrative comments could include:

a. Ability to work with others;

b. Ability to work under pressure;

c. Written and verbal communication skills;

d. Organizational skills; and

e. Leadership ability.

22. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL FLEET SCHOOL PACIFIC

PO 001 Assess Ship's Stability

1. Performance. Assess Ship's Stability.

2. Conditions.

   a. Given:

      (1) Manual of Trim and Stability (MT&S);

      (2) BDCS;

      (3) DWAN Computer access;

      (4) Operational ship;

      (5) GLM;

      (6) Personnel to make adjustments/readings;

      (7) Assistance;

      (8) flexible curve rulers;

      (9) Scientific calculators; and

      (10) Formula sheets.

   b. Denied: Nil.

   c. Environmental:

      (1) Onboard ship at sea or alongside;

      (2) In any weather condition; and

      (3) Day or night.

3. Standard. The Marine Technician Mechanical Supervisor shall assess ship's stability (WT0040) by:

   a. (**Med**) Identifying Stability States and forces acting on a ship;

   b. (**Med**) Extracting information from hydrostatic curves;

   c. (**Med**) Calculating the mean draught of a ship following a change of weight;

   d. (**Med**) Calculating ship's centre of gravity;

   e. (**Med**) Calculating the free surface effect in a ship's compartment;

   f. (**Med**) Calculating the rise in gravity due to Free Communication Effect (FCE); and

   g. (**Med**) Calculating stability following abnormal conditions.

4. References. A1, A9.

5. Task, Skill, Knowledge Numbers. WT0030, WT0040, WT0050, WT0080, WT0010, WT0110, WT0120.

6. Training Limitations. Nil.

PO 001 Assess Ship's Stability

EO 001.01 Identify stability states and forces acting on a ship

1. Performance. Identify stability states and forces acting on a ship.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Scientific calculators;

      (3) Formula sheets;

      (4) Stability models;

      (5) Animations/videos;

      (6) DWAN computers;

      (7) Stability extra practice booklet; and

      (8) Projectors.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician Mechanical Supervisor shall independently identify stability states and forces acting on a ship by:

   a. (Med) Investigating ship's stability (WT0050);

   b. (Med) Applying the stability factors;

   c. (Med) Monitoring ship's stability (WT0090);

   d. (Med) Utilizing the Manual of Trim and Stability; and

   e. (Med) Preparing for a stability trial and incline experiment.

4. Time Required. EO – 650 mins to deliver training

5. Remarks.

   a. Schedulers to liaise with Nav Arch at earliest opportunity to book guest speaker to speak with course on stability trials and incline experiments;

   b. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding of stability calculations;

   c. Trainees will begin creating formula sheets which they will add to throughout the duration of this EO; and

d. NWO have stability videos that may be used for use in the instruction of this PO. They will need to be reviewed by developers to determine if they are suitable for instruction.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Investigate ship's stability (WT0050)			
1. Explaining stability terminology IAW MT&S: a. Angle of Vanishing Stability; b. Archimedes Principle; c. Centre of Flotation; d. Displacement; e. Draught; f. Dynamic Stability; g. Equilibrium; h. Initial stability; i. Moment; j. Moment to Change Trim 1 cm; k. Period of a Ship; l. Perpendiculars; m. Righting Lever; n. Stiff Ship; o. Tender Ship; and p. Tonnes per Centimetre Immersion. 2. Explaining states of equilibrium; and 3. Explaining forces acting on a ship.	A9	**Method / Media:** F2F interactive lecture **Trg Environment:** Classroom **Activity Details:** The instructor will present the content to the class using videos, animations or models which demonstrate the effect these factors have on a ship's stability. **Trg Resources:** • PP&S; • Models/animations or videos of stability factors; • DWAN computer access; • Projector.	100
LESSON OBJECTIVE: Explain the theoretical stability factors			
1. Metacentre and metacentric height; 2. Keel to Metacentric height (KM); 3. Righting arm (GZ); 4. Upsetting arms; 5. Centre of Buoyancy; 6. Centre of Gravity; and 7. Relationship between stability factors in paras 1-5.	A9	**Method / Media:** F2F interactive lecture **Trg Environment:** Classroom **Activity Details:** The instructor will present the stability factors to the class using videos, animations or models which demonstrate the effect these factors have on a ship's stability. **Trg Resources:** • PP&S; • Models/animations or videos of stability factors; • DWAN computer access; • Projector.	200
LESSON OBJECTIVE: Monitor ship's stability (WT0090)			
1. List; 2. Trim; 3. Loll; 4. FSE;	A9	**Method / Media:** F2F interactive lecture **Trg Environment:** Classroom	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
5. FCE; and 6. Period of roll.		**Activity Details:** The instructor will present the stability factors to the class using videos, animations or models which demonstrate the effect these factors have on a ship's stability. **Trg Resources:** • PP&S; • Models/animations or videos of stability factors; • DWAN computer access; • Projector.	
LESSON OBJECTIVE: Utilize the Manual of Trim and Stability (MT&S)			
1. Purpose of the Manual of Trim and Stability. 2. Explain the sections of the (MT&S). 3. Using scenarios within the (MT&S).	A9	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** The instructor will present the MT&S and explain the sections of the manual. A demo will follow on how to use the manual to assist in assessing stability. Trainees will then be given a worksheet to complete that is based on the scenarios in the manual. Questions require the trainee to find information in the manual and answer general stability questions. Questions should be created so that they take trainees on a comprehensive "tour" of the manual to fully understand its contents and uses so they can use it to calculate and assess stability. **Trg Resources:** • PP&S; • MT&S for each trainee; • Activity worksheet; • DWAN computer access; • Projector.	200
LESSON OBJECTIVE: Preparing for a stability trial and incline experiment			
1. Describe trial ship's stability by identifying the procedures for an Incline Experiment (WT012). 2. Situations in which an Incline Experiment is conducted. 3. Requirement from Ship's Staff to conduct an Inclining Experiment. 4. Outcomes of an Incline Experiment.	A9	**Method / Media:** F2F guest speaker Nav Arch **Trg Environment:** Classroom **Activity Details:** The Nav Arch guest speaker will cover all TPts. **Trg Resources:** • PP&S;	50 min (guest speaker)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Models/animations or videos of stability factors • DWAN computer access • projector	

7. Test Details. There is no EC for EO 001.01. Trainees will be given ongoing feedback as they progress through this EO.

PO 001 Assess Ship's Stability

EO 001.02 Extract Information from hydrostatic curves

1. Performance. Extract information from hydrostatic curves.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Hydrostatic curves;

      (3) Scientific calculators; and

      (4) Formula sheets;

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician Mechanical Supervisor shall independently extract information from hydrostatic curves by:

   a. Analyzing ship's stability data.

   b. Assessing impact of hydrostatic curve on stability; and

   c. Experimenting with load adjustments to determine impact on hydrostatic curve.

4. Time Required. EO – 400 mins to deliver training

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO for use as they progress through each EO to further develop their understanding of stability calculations.

6. Teaching Points.

Teaching Points	**Ref**	**Method / Media / Environment / Training Activity Details**	**Time (mins)**
LESSON OBJECTIVES: Explain the purpose of the hydrostatic curves			
1. Analyze ship's stability data (WT0030) to include: a. The graphical presentation of a vessel and how it is measured using a series of draught marks. b. Length between perpendiculars (LBP). c. Length between draught marks (LBM) (also referred to as LBD). 2. Explain common items found on a Hydrostatic curve to include: a. Displacement in salt water. b. Vertical centre of Buoyancy (VCB). c. Transverse metacentre.	A1	**Method / Media:** F2F EDIP **Trg Environment:** Classroom **Activity Details:** The instructor to explain the components of a hydrostatic curve and how it's used to obtain variables used to calculate stability. Using a scenario, the instructor will demo how to utilize a hydrostatic curve to record the 7 parameters of from a hydrostatic curve. Trainees will follow with instructor. Trainees will then be given a worksheet and will complete individually where they will record the 7 parameters from the scenarios Instructor will review and provide feedback.	250 E 50 D 75 I/P 25 Debrief

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
d. Tons per centimeter immersion (TPC). e. Moment to change trim (MCT). f. Longitudinal centre of buoyancy (LCB). g. Longitudinal centre of flotation (LCF).		**Trg Resources:** • Hydrostatic curves • MT&S • DWAN computer • Projector • 2 scenarios • Solutions • Given mean draught • Rulers	

7. Test Details. There is no EC for EO 001.02. Trainees will be given ongoing feedback as they progress through this EO.

PO 001 Assess Ship's Stability

EO 001.03 Calculate mean draught of a ship following a change of weight

1. Performance. Calculate mean draught of a ship following a change of weight.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Scientific calculators; and

      (3) Formula sheets.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician Mechanical Supervisor shall calculate the mean draught of a ship following a change of weight, by:

   a. (Med) Calculating the draught at LCF (EMD), to include the following:

      (1) Trim between Draughts (TBD);

      (2) Trim between Perpendiculars (TBP); and

      (3) Draught at any point of the hull.

   b. (Med) Determining the position of the new LCF following an addition or removal of weight, to include:

      (1) Monitoring ship's draught (WT0080); and

      (2) Changes to the underwater volume of the ship's hull.

   c. (Med) Calculating change of trim (C of T) between perpendiculars (TBP) and C of T between draughts (TBD) following a forward/aft shift in weight (already onboard).

   d. (Med) Calculating new draught marks resulting in a transit from salt to fresh water.

4. Time Required.

   a. EO – 800 mins to deliver training;

   b. EC – 90 mins (total for all trainees).

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding of stability calculations.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Calculate draught at LCF (EMD)			
1. Calculating the draught at LCF (EMD), to include: a. Trim between Draughts (TBD). b. Trim between Perpendiculars (TBP). c. Draught at any point of the hull, by: i. Utilizing a similar triangle method.	A1	**Method / Media:** F2F, EDIP, SBL, homework assignments **Trg Environment:** Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems as they progress through the work. Instructor will provide ongoing feedback to trainees. Trainees will have one homework, scenario-based assignment given on the first day which covers material for the entire EO. The assignment will not be handed in until completion of EO. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Hydrostatic curves; • Solutions to problems; and • Formula sheet.	200
LESSON OBJECTIVE: Determine the position of the new LCF following an addition or removal of a weight			
1. Explaining the position of the new LCF following an addition or removal of a weight, to include: a. Monitor ship's draught (WT0080). b. Changes of the underwater volume of the ship's hull, to include: i. An increase in displacement; ii. Change in the LCF; and iii. Change of trim. c. Parallel sink/parallel rise	A1	**Method / Media:** F2F, EDIP, SBL, homework assignments **Trg Environment:** Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems as they progress through the work. Instructor will provide ongoing feedback to trainees. Trainees will have one homework, scenario-based assignment given on the first day which covers material for the entire EO.	200

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		The assignment will not be handed in until completion of EO. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Hydrostatic curves; • Solutions to problems; and • Formula sheet.	
LESSON OBJECTIVE: Calculate change of trim between perpendiculars and C of T between draughts following a forward/aft shift in weight			
1. Calculating the change of trim (C of T) between perpendiculars (TBP) and C of T between draughts (TBD) following a forward/aft shift in weight (already onboard), to include: a. New LCF. b. Trimming moments. c. Change of Trim (C of T). d. Obtaining new LCF, TPC, MCT from hydrostatic curves.	A1	**Method / Media:** F2F, EDIP, SBL, homework assignments **Trg Environment:** Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems as they progress through the work. Instructor will provide ongoing feedback to trainees. Trainees will have one homework, scenario-based assignment given on the first day which covers material for the entire EO. The assignment will not be handed in until completion of EO. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Hydrostatic curves; • Solutions to problems; and • Formula sheet.	200
LESSON OBJECTIVE: Calculate new draught resulting in a transit from salt to fresh water			
1. Calculating the new draught marks resulting in a transit from salt to fresh water.	A1	**Method / Media:** F2F, EDIP, SBL, homework assignments	200

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Trg Environment: Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems as they progress through the work. Instructor will provide ongoing feedback to trainees. Trainees will have one homework, scenario-based assignment given on the first day which covers material for the entire EO. The assignment will not be handed in until completion of EO. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Hydrostatic curves; • Solutions to problems; and • Formula sheet.	

7. Test Details. EO 001.03 will be tested in EC 001.03 and PC 001 IAW table below and Annex K.

EC 001.03: Calculate mean draught of a ship following a change of weight

EC Time: 75 mins + 15 min debrief

ASSESSMENT ACTIVITY: The trainee will be given a written exam assessing trainees' ability to calculate mean draught in a scenario format based on adding and subtracting weights along the centre line (CL). Using a single scenario, trainees will have to calculate new LCF, trimming moments, C of T and parallel sink/rise. Information from hydrostatic curves will also be required to complete the calculations. The trainee must show all calculations.

Pass/Fail criteria: 60% pass mark. Partial marks awarded for procedure in calculations.

ASSESSMENT RESOURCES:

- Exam with scenario and questions;
- Solution;
- Scientific Calculator;
- Formula sheet;
- MT&S;
- Hydrostatic curves;
- PP&S; and
- Rulers.

IMPACT OF FAILURE: Remedial Training.

PO 001 Assess Ship's Stability

EO 001.04 Calculate the ship's centre of gravity

1. Performance. Calculate the ship's centre of gravity.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Hydrostatic curves;

      (3) Rulers;

      (4) Scientific calculators; and

      (5) Formula sheets.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician Mechanical Supervisor shall calculate ship's centre of gravity, by:

   a. (Med) Determining where the ship's centre of gravity lies, to include:

      (1) Determining the principles that allow for the creation of a static curve of stability.

   b. (Med)Determining the ship's centre of gravity, to include:

      (1) Identifying the effect on the centre of gravity following a change in load;

      (2) Calculating the vertical shift in the centre of gravity;

      (3) Calculating the transverse shift in a centre of gravity;

      (4) Calculating list;

      (5) Calculating the longitudinal shift of G; and

      (6) Plotting a static curve of stability.

   c. (Med) Determining required weight change to adjust ship's stability (WT0090) by:

      (1) Identifying an unfavorable G position;

      (2) Calculating the required weight and location to shift G; and

      (3) Recommending an appropriate action to correct an unfavorable G position.

4. Time Required.

a. EO – 1900 mins to deliver training; and

b. EC – 125 mins (total for all trainees).

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding of stability calculations.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Determine where the ship's centre of gravity lies			
1. Identifying the principles that allow for the creation of a static curve of stability, to include: a. Curve of Static Stability. b. Righting arms. c. angles of heel. d. Explaining that static curves are dependent on: i. Centre of Gravity; ii. Freeboard; and iii. Beam of the ship. e. Explaining the Cross-Curves of stability. f. Explaining moments, to include: i. Moments about K; and ii. Moments about G.		**Method / Media:** F2F Interactive Lecture **Trg Environment:** Classroom **Activity Details:** The instructor will present the content in an interactive format that intersperses explanations with videos/animations of each TPt and application to real life scenarios. Stability model will be used to reinforce the points. **Trg Resources:** • DWAN computer access; • Projector; • Videos/animations; • Stability model; and • Handouts of content.	200
LESSON OBJECTIVE: Determine the ship's centre of gravity			
1. Effect on the centre of gravity following a change in load, by: a. Determining a transverse weight change to include: i. Overview; ii. Calculating the transverse shift in the centre of gravity; and iii. Calculating list. b. Determining a longitudinal weight change to include: i. Overview; ii. Calculating longitudinal shift of G; and iii. Calculating trim. c. Determining a vertical weight change, to include:	A1	**Method/Media:** F2F, EDIP, homework assignments **Environment:** Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems. Trainees will complete in class work and homework for each of transverse, longitudinal and vertical weight changes. Instructor will provide ongoing feedback to trainees upon completion of each weight change type. Culmination of lesson will be calculating simultaneous multi-directional weight changes.	1300 (300 for sub para a, b, c and 400 for d)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
i. Overview; ii. Calculating vertical shift in the centre of gravity; and iii. Calculating parallel sink and rise. d. Determining multi-directional weight changes to include: i. Transverse, longitudinal, and vertical weight changes; and ii. Plotting a static curve of stability.		**Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Graph paper; • Hydrostatic curves; • Static curve of stability; • Flexible curve ruler per trainee; • Solutions to problems; and • Formula sheet.	
LESSON OBJECTIVE: Determine required weight change to adjust ship's stability (WT0090)			
1. Identifying an unfavorable G position, to include: a. Calculating the new position of G after a weight change. b. Comparing the new position of G with the MT&S to determine if it is unfavorable. 2. Calculating the required weight and location to shift G, to include: a. Calculating different locations to add weight onboard. b. Calculating different weights/different physical size of weights. c. Combining weight and location calculations to determine the best course of action. 3. Recommending an appropriate action to correct an unfavorable G position.	A1	**Method/Media:** F2F, EDIP, homework assignments. **Environment:** Classroom **Activity Details:** Instructor will deliver the teaching points through EDIP where sufficient demonstration of calculations will enable trainees to work through consecutively more challenging problems. Trainees will complete in class work and homework for each required action to correct scenarios presented in TP2. Instructor will provide ongoing feedback to trainees as they calculate methods to correct possible issues with the scenarios in TP2. Culmination of lesson will be calculating simultaneous multi-directional weight changes that will result in an unfavorable position of G. The trainees will have to calculate a solution to correct the unfavorable position of G and present the solution in a "recommending to command" format. **Trg Resources:** • DWAN computer access; • SMART board;	100 (TP 1) 300 (TP 2,3)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Graph paper; • Hydrostatic curves; • Static curve of stability; • Flexible curve ruler per trainee; • Solutions to problems; and • Formula sheet.	

7. Test Details. EO 001.04 will be tested in EC 001.04 and PC 001 IAW table below and Annex K.

EC 001.04: Calculate the ship's centre of gravity

EC Time: 100 mins (concurrent) + 25 min debrief

ASSESSMENT ACTIVITY: The trainee will be given a written exam assessing trainee's ability to calculate the new centre of gravity after a multi-directional change in weight scenario. Trainees will have to calculate Information from hydrostatic curves will also be required to complete the calculations and plot the new static curve of stability. The trainee must show all calculations.

Pass/fail criteria: 60% pass mark. Partial marks awarded for procedure in calculations.

ASSESSMENT RESOURCES:

- Exam with scenario and questions;
- Solution;
- Scientific Calculator;
- Formula sheet;
- MT&S;
- Flexible curve ruler;
- Graph paper;
- Hydrostatic curves;
- PP&S; and
- Rulers.

IMPACT OF FAILURE: Remedial Training.

PO 001 Assess Ship's Stability

EO 001.05 Calculate the free surface effect in a ship's compartment

1. Performance. Calculate the free surface effect in a ship's compartment.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Scientific calculators; and

      (3) Formula sheets.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician mechanical supervisor shall calculate the free surface effect in a ship's compartment, by:

   a. (**Med**) Determining free surface effect to include the following:

      (1) Metacentric height;

      (2) Impact to G;

      (3) Wind healing arm; and

      (4) Countermeasures.

   b. (**Med**) Calculating the free surface effect to include:

      (1) Calculating the rise in Centre of Gravity due to free surface effect;

      (2) Calculating period of roll; and

      (3) Plotting the free surface correction on the GZ Curve.

4. Time Required. EO – 300 mins to deliver training.

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Determining free surface effect			
1. Explaining free surface effect to include: a. Metacentric height. b. Impact to G.	A1	**A Method / Media:** F2F, Interactive Lecture EDIP, videos	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
c. Wind healing arm. d. Countermeasures to include: i. Ship Building; ii. Tank Control (fully pressed or empty); and iii. Compartmentalizing damaged areas.		**Trg Environment:** Classroom **Activity Details:** The instructor will begin by presenting the content in an interactive format that intersperses explanations with videos/animations of each TPt and application to real life scenarios. **Trg Resources:** • DWAN computer access; and • SMART board.	
1. Calculating rise of G due to FSE to include: a. Overview. b. Changes to Metacentric height. c. Changes to G. d. Wind healing arm. e. Period of roll. f. Calculate FSE. g. Calculate new rise in G. h. Plot the new rise in G on the GZ curve.	A1	**Method / Media:** F2F, Interactive Lecture EDIP, videos, lecturer, homework assignments **Trg Environment:** Classroom **Activity Details:** Through EDIP, the instructor will provide sufficient demonstration of calculations to enable trainees to work through consecutively more challenging problems. Trainees will complete in class work a homework assignment on FSE. Instructor will provide ongoing feedback to trainees. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Rulers; • Graph paper; • Hydrostatic curves; • Static curve of stability; • Flexible curve ruler per trainee; • Solutions to problems; and • Formula sheet.	250

7. Test Details. EO 001.05 will be tested in conjunction with EO 001.06 in EC 001.06 and PC 001 and Annex K.

PO 001 Assess Ship's Stability

EO 001.06 Calculate the rise in gravity due to Free Communication Effect (FCE)

1. Performance. Calculate the rise in Gravity (G) due to FCE.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Scientific calculators; and

      (3) Formula sheets.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall calculate the rise in G due to FCE, by:

   a. (Med) Calculating FCE within internal compartments; and

   b. (Med) Calculating FCE in compartments open to sea.

4. Time Required.

   a. EO – 400 mins to deliver training; and

   b. EC – 115 mins (total for all trainees).

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding of stability calculations.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Calculate FCE within internal compartments			
1. Calculating FCE within internal compartments to include: a. Overview. b. Impacts to G. c. Metacentric height. d. Calculating FCE. e. Countermeasures.	A1	**Method / Media:** F2F, Interactive Lecture EDIP, videos, animations, lecturer, homework assignments **Trg Environment:** Classroom **Activity Details:** The instructor will begin by presenting the content in an interactive format that intersperses explanations with videos/animations of each TPt and application to real life scenarios.	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Through EDIP the instructor will provide sufficient demonstration of calculations to enable trainees to work through consecutively more challenging problems. Trainees will complete in class work and a homework assignment on FCE. As consolidation, trainees will then complete an activity to determine the rise in G due to a combination of FCE and FSE. Instructor will provide ongoing feedback to trainees. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Homework assignment; • Handouts/worksheets; • Solutions to problems; and • Formula sheet with SG values.	
1. Determining FCE in compartments open to sea to include: a. Overview. b. Impacts to G. c. Metacentric height. d. Calculating FCE. e. Countermeasures.	A1	**Method / Media:** F2F, Interactive Lecture EDIP, videos, animations, lecturer, homework assignments **Trg Environment:** Classroom **Activity Details:** The instructor will begin by presenting the content in an interactive format that intersperses explanations with videos/animations of each TPt and application to real life scenarios. Through EDIP the instructor will provide sufficient demonstration of calculations to enable trainees to work through consecutively more challenging problems. Trainees will complete in class work and a homework assignment on FCE. As consolidation, trainees will then complete an activity to determine the rise in G due to a combination of FCE and FSE. Instructor will provide ongoing feedback to trainees. **Trg Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee;	150 100 consolidation activity

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Homework assignment; • Handouts/worksheets; • Solutions to problems; and • Formula sheet with SG values.	

7. Test Details. Eos 001.05 and 001.06 shall be tested in conjunction in EC 001.06 and PC 001 IAW table below and Annex K.

EC 001.06: Calculate rise in G due to FSE and FCE

EC Time: 100 mins + 15 min debrief

ASSESSMENT ACTIVITY: The trainee will be given a written exam to calculate the rise in G due to a combination of FSE and FCE. One scenario will encompass both FSE and FCE simultaneously. Trainees will complete separate calculations for each and then use those solutions to determine overall rise in G. Trainees must show all calculations.

Pass/fail criteria: Partial marks awarded for procedure in calculations. 60% pass mark

ASSESSMENT RESOURCES:

- Exam;
- Solution;
- Scientific Calculator;
- Formula sheet with SG values; and
- MT&S.

IMPACT OF FAILURE: Remedial Training.

PO 001 Assess Ship's Stability

EO 001.07 Calculate stability following an abnormal condition

1. Performance. Calculate stability following an abnormal condition.

2. Conditions.

   a. Given:

      (1) MT&S;

      (2) Scientific calculators; and

      (3) Formula sheets.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The Marine Technician Mechanical Supervisor shall calculate stability following an abnormal condition to include:

   a. (**Med**) Determining the effects of icing on a ship by:

      (1) Identifying the effects of icing on a ship's characteristics;

      (2) Identifying factors affecting the formation of ice;

      (3) Determining methods of ice removal;

      (4) Determining factors that affect the weight of ice;

      (5) Calculating the rise of the centre of gravity of a ship, using the icing tables; and

      (6) Calculating the trimming moment, using the icing tables.

   b. (Med) Calculating GM when a ship is grounded to include:

      (1) Identifying possible grounding incidents;

      (2) Calculating GM when it's grounded to include:

         i. The full length of its keel;

         ii. When grounded at a point on the CL; and

         iii. When grounded at a point off the CL.

   c. (Med) Adjusting Ship's Stability following an abnormal event.

4. Time Required.

   a. EO – 950 mins to deliver training;

   b. EC – 115 mins (total for all trainees); and.

   c. PC –210 mins (total for all trainees).

5. Remarks. Trainees should be given the Stability Extra Practice Booklet at the start of this PO so they can use it as they progress through each EO to further develop their understanding of stability calculations.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Determining the effects of icing on a ship			
1. Effects of icing on a ship's characteristics to include: a. Stability; b. List and Trim; c. Freeboard; d. Maneuvering and control; e. Equipment; and f. Safety. 2. Factors affecting the formation of ice, to include: a. Air and sea temperatures; b. Wind speed and direction; c. Displacement and freeboard; d. Sea state; e. Precipitation; and f. Time of day. 3. Methods of ice removal from ships to include: a. Physical removal to include: i. Hand tools; and ii. Steam de-icing. b. Course alteration to include: i. Wind correction; ii. Speed correction; and iii. Warmer climate. 4. Factors that determine the weight of ice, to include: a. Thickness. b. Density. c. Utilizing the icing tables in the MT&S. 5. Calculating the rise of the centre of gravity of a ship, using icing tables. 6. Calculating the trimming moment, using icing tables.	A1	**Method / Media:** F2F Guided Discussions/activity, case study guides. **Trg Environment:** Classroom **Activity Details:** Trainees will be given a case study of an icing event the day before to read. Using the case study as a guide, the instructor will lead a guided discussion/activity with the class on the effects of icing, the calculations and, counter measures to combat icing. Questions should cover all material in the TPts. Using the case study, the class will calculate the change in G due to icing. Instructor will facilitate the discussion, provide feedback, cover any points missed or not fully elaborated on by trainees. **Trg Resources:** • Case study on an icing event; • guided discussion questions • expected responses; • MT&S; • Formula sheets; • Scientific calculators; • Hydrostatic curves; and • Flip chart and paper if required.	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Calculating GM when a ship is grounded			
1. Identifying possible grounding incidents: a. Beaching. b. Broaching. c. Being driven further ashore. d. Pounding on the ocean bottom. e. Suction effect of being grounded into the seabed (mud). f. Onset of flooding. 2. Calculating GM when a ship is grounded to include: a. The full length of its keel. b. When grounded at a point on the CL. c. When grounded at a point off the CL.	A1	**Method / Media:** F2F, EDIP, videos, animations. **Trg Environment:** Classroom **Activity Details:** The instructor will introduce grounding as it relates to calculating stability. This will include the use of videos/animations or images depicting each type of incident. Instructor will provide sufficient demonstration of calculations to allow trainees to work through consecutively more challenging problems. **Training Resources:** • DWAN computer access; • Animations/videos (TEL); • SMART board; • Scientific calculators; • MT&S for each trainee; • Handouts/worksheets; • Solutions to problems; and • Formula sheet.	E 50 DI 200 P 200
LESSON OBJECTIVE: Adjust Ship's Stability following an abnormal event			
Method/Media: F2F Instructor Led Group Activity, SBL, scenario materials **Environment:** Classroom **Activity Details:** This activity is the culmination of all EOs within this PO. Now that they can complete all calculations trainees can make a recommendation to adjust ship's stability. The instructor will use a scenario where a ship experiences an abnormal sailing event which could include icing and/or a grounding event. This results in FSE, FCE, and other changes in weight which require completion of all calculations performed within PO 001 to assess the ship's stability. The instructor will lead trainees through the scenario and ask trainees to identify the events occurring and the required calculations needed as a result. Trainees will individually complete the calculation and instructor will confirm before moving to the next calculation in the scenario. Upon completion of the calculations, trainees will recommend adjustments to ship's stability to the instructor. **Training Resources:** • DWAN computer access; • SMART board; • MT&S for each trainee; • Handouts/worksheets; • Rulers;			400

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
• Graph paper; • Scientific calculator; • Hydrostatic curves; • Static curve of stability; • Flexible curve ruler per trainee; • Solutions to problems; and • Formula sheet with SG values.			

7. Test Details. EO 001.07 will be tested in EC 001.07 and PC 001 IAW table below and Annex K.

EC 001.07: Calculate a ship's stability due to an abnormal condition

EC Time: 100 mins + 15 min debrief

ASSESSMENT ACTIVITY: The trainee will be given a written exam and will answer scenario-based questions on icing and grounding events.

Icing questions will require trainees to obtain information from the MT&S based on an icing scenario.
Grounding questions will consist of required calculations to determine new GM based on the scenario.

The trainee will be required to show all calculations.

Pass/fail criteria: Partial marks awarded for procedure in calculations. 60% pass mark

ASSESSMENT RESOURCES:

- Exam with scenario;
- Solution;
- Scientific Calculator;
- Hydrostatic curves;
- Rulers;
- Formula sheet; and
- Manual of Trim and Stability.

IMPACT OF FAILURE: Remedial Training.

PC 001: Assess ship's stability

PC Time: 180 mins + 30 min debrief

ASSESSMENT ACTIVITY: This will be a written exam which is the culmination of all EOs within this PO. The test will be scenario based where a ship experiences an abnormal sailing event which will include: FSE; FCE; load change due to icing; internal load shift; and grounding. This will require completion of all calculations performed within PO 001 to assess the ship's stability. Trainees will individually calculate the new G and then graph it on a GZ curve.

The trainee will be required to show all calculations.

Pass/fail criteria: Partial marks awarded for procedure in calculations. 60% pass mark

ASSESSMENT RESOURCES:

- Exam with scenarios;
- Solution;
- Hydrostatic curves;
- Flexible curve ruler;
- Ruler;
- Static curve of stability;
- Graph paper;
- Scientific Calculator;
- Formula sheet with SG values; and
- Manual of Trim and Stability.

IMPACT OF FAILURE: Re-test / TRB IAW Ch 2 and NPTORD

PO 002 Perform the Duties of Fuelling I/C.

1. Performance. Perform the duties of fuelling I/C.

2. Conditions.

   a. Given:

      (1) Operational ship with fuelling capabilities;

      (2) Qualified fuelling party;

      (3) References;

      (4) Communications equipment;

      (5) Supervision;

      (6) Keys to restricted spaces;

      (7) Fuel spill response kits;

      (8) HAZMAT containment capabilities;

      (9) Personal PPE;

      (10) Fuel quality test kits;

      (11) Fuel (DFO and JP5); and

      (12) Updated fuel report.

   b. Denied: Nil.

   c. Environmental: Onboard ship, at sea or alongside, in any condition.

3. Standard. The Marine Technician Mechanical Supervisor shall independently perform the duties of the fuelling I/C IAW refs by:

   a. (**Low**) Initiating fuelling orders to include:

      (1) Completing fueling memo; and

      (2) Completing fuelling orders.

   b. (**High**) Leading maritime vessel fueling (internal fuel/ providing/ consolidation) (OT0090) to include:

      (1) Preparing a ship to move (embark /disembark / transfer) fuel;

      (2) Moving (embark / disembark / transfer) fuel; and

      (3) Securing a ship from fuelling operations.

   c. (**Low**) Finalizing fuelling orders.

4. References. A2, A3, A4, A5.

5. Task, Skill, Knowledge Numbers. OT0090, OT0010.

6. Training Limitations. To instruct effectively, the TE requires an operational system control interface (IPMS/IPCS).

PO 002 Perform the Duties of Fuelling I/C

EO 002.01 Initiate fuelling orders

1. Performance. Initiate fuelling orders.

2. Conditions.

   a. Given:

      (1) Classroom;

      (2) PP&S;

      (3) Computer with DWAN access;

      (4) References; and

      (5) Scenarios with fuel spec sheets.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall independently initiate fuelling orders, IAW refs by:

   a. (**Low**) Advising command on fuel quality (OT0010) to include:

      (1) Analyzing fuel test results against POL handbook;

      (2) Identifying substitutions for non-standard fuels; and

      (3) Formulating fueling recommendations.

   b. (**Low**) Adhering to SEMS SOPs by:

      (1) Detailing fueling party duties;

      (2) Determining the type of fueling evolution; and

      (3) Selecting fueling checklists based on the fuelling evolution.

   c. (**Low**) Completing fueling documentation to include:

      (1) Extracting data from fuel reports;

      (2) Initiating fueling orders prior to fuelling evolutions;

      (3) Completing fueling memo; and

      (4) Finalizing fueling orders after fuelling evolutions.

4. Time Required.

   a. EO – 120 mins to deliver training; and

   b. EC – 65 mins (total for all trainees).

5. Remarks. Trainees have experience with SEMS SOP E1 and participating in fuelling evolutions in various positions. The purpose of this EO is to activate trainee experience and direct it specifically to applying SEMS in the completion of the checklist, not completing the checklist itself.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE: Advise command on fuel quality			
1. Advise command on fuel quality (OT0010): a. Analyzing fuel test results against POL handbook: i. DFTK results; ii. Clear and bright test; iii. Review vendor spec sheets; and iv. POL handbook. b. Identifying substitutions for non-conforming fuels: i. North American fuels; ii. Non North American fuels; iii. Acceptable substitutions IAW refs; and iv. Implications of embarking non-conforming fuels. c. Formulating fuelling recommendations: i. Fuel Decision Tree (Tech Bulletin); ii. Recommendation options; and iii. Impacts to recommendations.	A3 A2SOP E1 A4 A5	**Method / Media:** F2F GD **Trg Environment:** Classroom **Activity Details:** The instructor will lead a guided discussion on advising command. Trainees collectively have the knowledge and experience on these TPts and the guided discussion will allow trainee's experience to form the basis of instruction. Instructor will amplify important safety or points not addressed in trainee discussion. Questions for discussion should focus on: • Fuel quality; • Fuel types; • References and how to use them, specifically SEMS SOP E1, POL Handbook and Tech bulletins; • Fuelling party positions trainees have held and important points of those positions; • Key fuelling positions like the fuelling I/C and what those duties include; and • Fuelling documentation. **Trg Resources:** Guided Discussion: • instructions; and • Prepared guided discussion notes and questions.	GD 20
LEARNING OBJECTIVE: Adhering to SEMS SOPs			
1. Adhering to SEMS SOPs: a. Describing the duties of the fueling party IAW SEMS SOP E1 to include: i. Positions; and ii. Responsibilities. b. Explaining the types of fueling evolutions: i. Alongside; ii. RAS;	A3 A2SOP E1 A4 A5	**Method / Media:** F2F/Guided Discussion (GD) **Trg Environment:** Classroom **Activity Details:** The instructor will lead a guided discussion SEMS SOP E1. Trainees collectively have the knowledge and experience on these TPts and the guided discussion will allow trainee's experience to form the basis of instruction.	GD 20

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
iii. TAC RAS; iv. Transferring fuel; v. De-fuelling; vi. Helicopter fueling; and vii. Fueling small craft. c. Describing fueling checklists: i. Fueling party; ii. Pre-fuelling; iii. Manual dip verification checklist to include: (1) Pre-fuelling; and (2) Post-fuelling. iv. Valve operation; v. Post-fuelling; and vi. Fueling report.		Instructor will amplify important safety or points not addressed in trainee discussion. Questions for discussion should focus on: • Fuel quality; • Fuel types; • References and how to use them, specifically SEMS SOP E1, POL Handbook and Tech bulletins; • Fuelling party positions trainees have held and important points of those positions; • Key fuelling positions like the fuelling I/C and what those duties include; and • Fuelling documentation. **Trg Resources:** Guided Discussion: • instructions; and Prepared guided discussion notes and questions.	
LEARNING OBJECTIVE: Completing fuelling documentation			
1. Completing fuelling documentation: a. Extracting data from fuel reports: i. Before and after fuelling; ii. Calculating fuel requirements; and iii. Calculate consolidation requirements. b. Initiating fuelling documentation prior to fuelling evolutions: i. Assigning personnel to fuelling party; ii. Determining tank filling sequence; and iii. Determining time requirements. c. Completing fueling memo. d. Finalizing fuelling documentation after fuelling evolutions: i. Confirming amount of fuel received; ii. Submitting paperwork to MSEO.	A3 A2SOP E1 A4 A5	**Method / Media:** GD and assignment **Trg Environment:** Classroom **Activity Details:** The instructor will lead a guided discussion on completing fuelling documentation. Next, trainees will be given 20 scenario-based questions and apply policy from SEMS SOP E1 in their answers. Questions will consist of questions on fuel quality, fuelling evolutions, calculating fuel quantity, fuel tank sequencing and assigning personnel to fuelling party positions. Scenarios should include alongside, at sea and transfer operations. **Fuelling Orders Activity:** • Scenarios; • Calculator; • Access to references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree); • DWAN computer access; • Fuel spec sheets (that meet or don't meet spec); • List of personnel for scheduling fuelling party; • PP&S as required.	GD 20 Assign 60

7. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002 IAW table below and Annex K.

EC 002.01: Complete Fuelling Orders

EC Time: 50 min +15 min for debrief

A**ssessment Activity**: Trainees will be given a copy of a fuelling order based on a given scenario and they are to review it for accuracy and completeness. Errors and omissions will be inserted into the order and trainees will identify and correct errors and fill in any missing information. Errors and omissions should include fuel quality, fuelling evolutions, tank sequencing, and assigning personnel.

Pass/Fail criteria: Pass/Fail 70% with feedback provided by instructor.

Assessment Resources:

- Scenarios;
- Calculator;
- Access to references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree);
- DWAN computer access;
- Fuel spec sheets (that meet or don't meet spec);
- List of personnel for scheduling fuelling party;
- PP&S as required.

IMPACT OF FAILURE: Remedial Training.

PO 002 Perform the duties of Fuelling I/C

EO 002.02 Lead Maritime Vessel Fuelling

1. Performance. Lead Maritime Vessel Fuelling (internal fuel / providing / consolidating) (OT0090).

2. Conditions.

   a. Given:

      (1) BDCS trainer;

      (2) Role players as Watch-keepers and fuelling party;

      (3) Communications equipment;

      (4) Classroom;

      (5) PP&S;

      (6) Computer with DWAN access;

      (7) References; and

      (8) Fuelling scenarios;

   b. Denied: Assistance.

   c. Environmental: Classroom and BDCS trainer.

3. Standard. The trainee shall independently lead maritime vessel fuelling IAW refs by:

   a. Preparing a ship to move (embark / disembark / transfer) fuel to include (NT):

      (1) Consolidating fuel tanks;

      (2) Briefing the fuel party; and

      (3) Configuring fuel system to receive fuel.

   b. (**High**) Moving (embark / disembark / transfer) fuel to include:

      (1) Initiating fuelling evolutions;

      (2) Monitoring flow rates;

      (3) Filling tanks IAW tank filling sequence; and

      (4) Stopping fuelling evolutions when tanks do not exceed:

         (a) 95% capacity alongside; or

         (b) 90% at sea.

   c. Securing a ship from fuelling operations to include (NT):

(1) Reconfiguring fuel systems;

(2) Securing fuelling equipment; and

(3) Securing personnel.

4. Time Required.

a. EO – 90 mins to deliver training;

b. EC – 600 mins (50 min test and 10 min debrief per trainee); and

c. PC – 500 mins (50 mins test and 10 mins debrief per trainee).

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LEARNING OBJECTIVE: Preparing a ship to move (embark / disembark / transfer) fuel			
1. Preparing a ship to move (embark / disembark / transfer) fuel to include (NT): a. Consolidating fuel tanks; b. Briefing the fuel party; and c. Configuring fuel system to receive fuel.	A2 SOP E1	**Method / Media:** Small group training activity **Trg Environment:** Classroom **Activity Details:** In groups, trainees will prepare and present a brief to the class based on the following parts of a fuelling evolution: • Initiating fuelling evolutions; • Monitoring flow rates; and • Filling tanks; and • Stopping fuelling evolutions. Briefings must include the teaching points and references associated with fuelling. Trainees should include real examples where possible. **Trg Resources:** • DWAN computer access; • Projector; • References; Flipchart and paper.	30 min develop 40 min present (10 min/gp) 20 min debrief (5 min/gp)
LEARNING OBJECTIVE: Lead maritime vessel fuelling			
1. Moving (embark / disembark / transfer) fuel to include: a. Initiating fuelling evolutions: i. Dump set up: (1) Drip trays; (2) Fuel spill kits; and	A2 SOP E1	Combined with LO above	Combined with above

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
(3) Tools. ii. Communications checks: (1) Tank sentries; (2) Valve sentries; and (3) Dump personnel. iii. Tank filling: (1) operating locally and remotely; (2) valve operation order. iv. Path of orders to start/stop pumping: (1) Inform command ready to embark fuel. b. Monitoring flow rates: i. Confirming fuel is embarking by monitoring change in levels; and ii. Calculate rate of flow using change in volume over time. c. Filling tanks IAW tank filling sequence to include: i. Ship's list; ii. Tank fill and stop timings based on the fuel rate; iii. operating locally and remotely; iv. valve operation order; v. Opening and closing tank valves; vi. Tank levels in isolated tanks; and vii. Manual dips. d. Maximum tank capacity: i. Maximum tank capacity: (1) At sea; and (2) Alongside. ii. Fuelling warnings: (1) 10 min; (2) 5 min; (3) 3 min; (4) 1 min; (5) Standby; and (6) Stop pumping. iii. Stopping pumping to include:			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
(1) Maximum capacities alongside and at sea; (2) Tanks filled during normal fuelling; (3) Emergencies; (4) Blow down/gravity drain; and (5) Closing tank valves remotely and locally.			

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002 IAW table below and Annex K.

EC 002.02: Lead Vessel Fuelling

EC Time: 50 min per trainee + 10 min for debrief per trainee (600 min total)

ASSESSMENT ACTIVITY: The trainee will be given scenario with current fuel load the day prior to EC. The scenario will be the same for all trainees. The trainee will calculate the amount of fuel required, consolidation plan and determine filling sequence, which the instructor will input into BDCS. Trainee will brief fuelling party IAW SEMS.

Scenario will have trainees start fuelling evolution and fill three tanks. During filling there will be an abnormal event and the trainee will be required to stop and explain the event, why it happened, and how to correct to instructor to continue fuelling.

At least one tank must be filled prior to abnormal event. Trainee will brief instructor on amount of fuel received. Trainees must not exceed filling levels detailed in checklist.

Pass/fail criteria: Pass/Fail 70% IAW EC 002.02 checklist

ASSESSMENT RESOURCES:

- Scenario handed out the day prior;
- Access to references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree);
- DWAN computer access;
- List of personnel for scheduling fuelling party;
- BDCS trainer;
- Communications equipment;
- Calculators;
- Clock/stopwatch; and
- Standard fuel brief template.

IMPACT OF FAILURE: Remedial Training.

PO2 MECHANICAL SUPERVISOR – EC 002.02 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with asterisk *) and achieve 70% SATs on the remainder of the checklist items to pass this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Prepare a ship to move (embark / disembark / transfer) fuel			
Provide recommendation to consolidate fuel tanks			
Brief the fuel party IAW template			
2. Move (embark / disembark / transfer) fuel			
Confirms inspection of dump to include: • Drip trays • Fuel spill kits • Tools			
Perform communications checks with: • Tank sentries • Valve sentries • Dump personnel			
Direct opening and closing tank filling valves locally and remotely			
Inform command ready to embark fuel			
3. Monitoring Flow Rate			
Confirms fuel is embarking by monitoring change in levels on BDCS			
Calculate rate of flow using change in volume over time • Direct entries in event log			
4. Filling tanks IAW tank filling sequence to include:			
Monitor ship's list by asking assessor for inclinometer readings			
Calculate tank fill and stop timings using fuel rate			
* Direct opening and closing tank valves as tanks fill by opening one before closing the previous one			
Monitor tank levels in isolated tanks to ensure levels aren't rising			
Confirm tank levels through manual dips • Dispatch tank sentries to verify readings on BDCS			
Correct actions taken when an abnormal event occurs • *Stop fuelling • Based on abnormal event, explains to assessor:			

ITEM	SAT	UNSAT	COMMENTS
○ the event ○ why fuelling was stopped ○ how to correct event ○ next actions after event corrected			
Makes appropriate recommendations in order to resume fuelling			
5. Stopping Fuelling Evolutions			
Announce fuelling warnings at:			
• 10 min;			
• 5 min;			
• 3 min;			
• 1 min;			
• Standby;			
• *Stop pumping.			
Stop pumping to include:			
• * No tanks exceed 95% alongside or 90% at sea;			
• Direct closing of tank valves remotely and locally.			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002: Lead Maritime Vessel Fuelling

PC Time: 40 min per trainee + 10 min for debrief per trainee (500 min total)

ASSESSMENT ACTIVITY: The trainee will be given scenario with current fuel load the day prior to EC. The scenario will be the same for all trainees. The trainee will calculate the amount of fuel required, consolidation plan and determine filling sequence, which the instructor will input into BDCS.

Trainee will brief fuelling party IAW SEMS. Scenario will have trainees start fuelling evolution and fill three tanks. Trainee will brief instructor on amount of fuel received. Trainees must not exceed filling levels detailed in checklist.

Pass/fail criteria: Pass/Fail 70% IAW PC 002 checklist

ASSESSMENT RESOURCES:

- Scenario handed out the day prior;
- Access to references (SEMS SOP E1, Tech Bulletin, POL Handbook, Fuel Decision Tree);
- DWAN computer access;
- List of personnel for scheduling fuelling party;
- BDCS trainer;
- Communications equipment;
- Calculators;
- Clock/stopwatch; and
- Standard fuel brief template.

IMPACT OF FAILURE: Re-test / TRB IAW Ch 2 and NPTORD

PO2 MECHANICAL SUPERVISOR – PC 002 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with asterisk *) and achieve 70% SATs on the remainder of the checklist items to pass this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Prepare a ship to move (embark / disembark / transfer) fuel			
Provide recommendation to consolidate fuel tanks			
Brief the fuel party IAW template			
2. Move (embark / disembark / transfer) fuel			
Confirms inspection of dump to include: • Drip trays • Fuel spill kits • Tools			
Perform communications checks with: • Tank sentries • Valve sentries • Dump personnel			
Direct opening and closing tank filling valves locally and remotely			
Inform command ready to embark fuel			
3. Monitoring Flow Rate			
Confirms fuel is embarking by monitoring change in levels on BDCS			
Calculate rate of flow using change in volume over time • Direct entries in event log			
4. Filling tanks IAW tank filling sequence to include:			
Monitor ship's list by asking assessor for inclinometer readings			
Calculate tank fill and stop timings using fuel rate			
* Direct opening and closing tank valves as tanks fill by opening one before closing the previous one			
Monitor tank levels in isolated tanks to ensure levels aren't rising			
Confirm tank levels through manual dips • Dispatch tank sentries to verify readings on BDCS			
Correct actions taken when an abnormal event occurs • *Stop fuelling • Based on abnormal event, explains to assessor:			

ITEM	SAT	UNSAT	COMMENTS
○ the event ○ why fuelling was stopped ○ how to correct event ○ next actions after event corrected			
Makes appropriate recommendations in order to resume fuelling			
5. Stopping Fuelling Evolutions			
Announce fuelling warnings at:			
• 10 min;			
• 5 min;			
• 3 min;			
• 1 min;			
• Standby;			
• *Stop pumping.			
Stop pumping to include:			
• * No tanks exceed 95% alongside or 90% at sea;			
• Direct closing of tank valves remotely and locally.			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 Manage Battle Damage Control Organization IT0350

1. Performance. Manage Battle Damage Control Organization (IT0350).

2. Conditions.

   a. Given:

      (1) References;

      (2) Supervision;

      (3) BDCS/Incident board;

      (4) DC org;

      (5) Communications; and

      (6) General Load Monitor (GLM).

   b. Denied: Nil.

   c. Environmental: Ship at Sea/Alongside in any situation and under any condition.

3. Standard. The Marine Technician Mechanical Supervisor shall independently manage battle damage control organization IAW SSOs by:

   a. (**High**) Performing Damage Control Petty Officer (DCPO) duties by:

      (1) Conducting Initial Actions (IA);

      (2) Utilizing DC resources;

      (3) Directing SBT I/Cs;

      (4) Briefing Command on DC considerations; and

      (5) (NT) Maintaining control of BDCS to include directing the HQ1 plotter.

   b. Submitting a post-incident report by:

      (1) Carrying out a preliminary assessment following an incident; and

      (2) Drafting a post-incident report.

4. References. A1, A2, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15.

5. Task Numbers. AFT1520; IT0120; IT0130; IT0810, IT0350.

6. Training Limitations. DCTF is the preferred location of practical training as a BDCS trainer is required and DCTF has more realism for HQ1. MT&S is CTAT so development decisions must take this into consideration.

PO 003 Manage Battle Damage Control Organization

EO 003.01 Perform DCPO duties

1. Performance. Perform DCPO duties

2. Conditions.

   a. Given:

      (1) DWAN Workstation;

      (2) BDCS trainer (DCTF preferred);

      (3) DC scenarios;

      (4) Primary and alternate comms (telephones, PRCs);

      (5) Applicable risk folio;

      (6) Supervision;

      (7) Assistance; and

      (8) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) BDCS Trainer (At DCTF preferred).

3. Standard. The trainee shall perform DCPO duties IAW current SSOs and CFCD 132 by:

   a. (**High**) Conducting initial actions by:

      (1) Establishing communications with all DC stations;

      (2) Identifying ship's status;

      (3) Identifying the location(s) of all damage;

      (4) Assessing the damage; and

      (5) Reporting the DC Organization is closed up to command in under 6 minutes.

   b. (**High**) Utilizing appropriate DC resources.

   c. (**High**) Directing SBT I/Cs by:

      (1) Assigning DC priorities;

      (2) Directing the HQ1 plotter to include controlling of BDCS;

(3) Monitoring the progress of the DC Organization on BDCS;

(4) Briefing command on DC considerations to include the following:

(a) Fire considerations;

(b) Flood considerations; and

(c) Smoke management.

(d) (**High**) Identifying options to maintain stability in emergencies;

(e) (**High**) Explaining post-event options in the event of grounding; and

(f) (**High**) Explaining post-event options in the event of a sustained/main machinery space fire IAW current CFCD 132 procedures.

4. Time Required.

a. EO – 1805 mins to deliver training; and

b. EC – 700 mins (total for all trainees).

5. Remarks. DCTF is the preferred location for practical training and assessments, but can use MS Div BDCS trainer if DCTF unavailable.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Manage battle damage control organization			
1. Command station: a. DC related tasks. b. DC related responsibilities. 2. DC HQ: a. Manning responsibilities i. DCO; ii. DCC operator (plotter); iii. Elec manager; iv. MCR; v. Containment monitor; and vi. ChEng. b. Tasks. 3. CS ERT HQ: a. Responsibilities. b. Capabilities. 4. Med HQ: a. Casualty clearing teams.	A6 A7 Part 1 Ch. 1 para 16-17	**Method / Media:** Self-Study, References **Trg Environment:** Classroom **Activity Details:** Trainees will be given SSOs and CFCD 132 as well as a question sheet (20 questions) and will have to find the answers to the questions in the references. Instructor to have a standardized solution for final review. **Trg Resources:** • References • Question sheet • Standardized solution	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
b. Locations. 5. RIT: a. Capabilities. b. Limitations. 6. RAT: a. Capabilities. b. Limitations. 7. Manning pool: a. Capabilities. b. Resources available.			
LESSON OBJECTIVE: Conduct Initial Actions			
1. Establish communication with all DC stations. 2. Identify ship's status. 3. Identify location of all damage. 4. Assess the damage by: a. Initiating rapid survey. b. Assigning SBT duties. c. Assign boundaries. d. Assign use of fitted systems. 5. Report the DC Org Closed up to command (IAW platform standard).		**Method / Media:** BDCS, SHINCOM EDI **Trg Environment:** DCTF preferred, BDCS trainer if DCTF unavailable **Activity Details:** Instructor to explain and demonstrate initial actions; the trainees will imitate once with instructor feedback. **Trg Resources:** • BDCS • SHINCOM (equivalent communications)	10 (ED) 50 (I)
LESSON OBJECTIVE: Identify appropriate DC resources			
1. Risk folio. 2. MT&S. 3. DCC Operator. 4. SBT recommendations. 5. Personnel. 6. Fitted systems (eg. Pre-wet). 7. Portable systems/equipment 8. Specialty tools (eg. PowerHawk extraction tool; Exothermic cutting torch). 9. BDCS. 10. Communications (SHINCOM).	A8 A7 A13	**Method / Media:** Brainstorm **Trg Environment:** Classroom **Activity Details:** Trainees to brainstorm appropriate resources; instructor to confirm they hit on all TPts. Trainees can discuss their own experiences with the various resources. **Trg Resources:** Risk folio	25
LESSON OBJECTIVE: Direct SBT I/Cs			
1. Assign priorities. 2. Direct BDCS operator/plotter 3. Monitor progress of DC org. 4. Confirm resources 5. Allocate personnel in support of the current priority to include: a. Fire, to include: i. galley/trunking fires; ii. ammunition and magazine fires; iii. flammable liquid fires;	A1 A7 Ch. 7 A13	**Method / Media:** GD **Trg Environment:** Classroom **Activity Details**: Instructor will ask trainees to discuss the methods of directing SBT I/Cs, asking pertinent questions to ensure all TPts are covered off. **Trg Resources:** NIL	120

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
iv. fires involving gas cylinders; v. fires involving electrical equipment; vi. flight deck fire fighting; and vii. fire fighting in an NBC environment. b. Flood/stability. c. Smoke management. d. Relocations.			
LESSON OBJECTIVE: Identify options to maintain stability in Emergencies			
1. Down-flooding. 2. Jettisoning based on Jettison Bill. 3. Counter-flooding. 4. Shift of CG. 5. Ballasting. 6. Pumping out. 7. Loss of a space to sea. 8. Sectioning. 9. Pressing a tank (reduce FSE).	A1 A10 A12 A15	**Method / Media:** Guided discussion **Trg Environment:** Classroom **Activity Details:** Instructor will ask trainees to discuss the methods of maintaining stability in emergencies, asking pertinent questions to ensure all TPts are covered off. **Trg Resources:** MT&S	40
LESSON OBJECTIVE: Explain HQ1's actions in event of grounding			
1. Effects of grounding on a ship's hull. 2. General considerations to be taken when a ship is grounded. 3. Identify extent of damage. 4. Type of grounding. 5. Manage the damage. 6. Explain post-event options: a. Remaining in place. b. Towed by accompanying assets. c. Self-extraction (eg. Re-ballast).	A7 Ch. 7 art 707.4 A9	**Method / Media:** GD **Trg Environment:** Classroom **Activity Details:** Instructor will ask trainees to discuss the actions taken during a grounding, asking pertinent questions to ensure all TPts are covered off. **Trg Resources:** • Workstation (DWAN) • CFCD 132 Ch. 7 art 707.4	60
LESSON OBJECTIVE: Explain HQ1's actions in event of sustained/main machinery space fires			
1. Identify resources: a. Risk folio. b. MT&S. c. BDCS Operator. d. SBT recommendations. e. Personnel.	A9 A7 Art 606 A13	**Method / Media:** Case Study **Trg Environment:** Classroom **Activity Details**: Instructor to present case studies based on major military events, (eg: HMCS OTTAWA GT fires 2003, HMCS PROTECTUER engine room fire 2014). Instructor to discuss current methods of fighting a sustained/main machinery space fire based on CFCD policy. **Trg Resources:** • CFCD 132 Art 606	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		• Risk folio • MT&S	
LESSON OBJECTIVE: Perform DCPO duties			
Method / Media: Practice, SBL **Trg Environment:** BDCS Trainer (DCTF Preferred) **Activity Details:** Trainees to complete a minimum of two practice runs as DCPO. First run simple, additional runs to be more complex. Trainees not acting as DCPO will act as SBT IC, Elec Mgr, ERT, Casualty clearer, and Manning Pool IC. **Trg Resources:** • DC boards (to match BDCS trainer); • BDCS trainer (DCTF preferred); • Primary / Alternate comms (telephones, PRCs); • DC-based scenarios; and • Role-players (candidates not presently being assessed).			60 per run 10 per debrief (1400 total)

7. Test Details. EO 003.01 will be tested in EC 003.01 and PC 003 IAW table below and Annex K.

EC 003.01: Perform DCPO duties

EC Time: 50 mins + 20 mins debrief per trainee (Total 700)

ASSESSMENT ACTIVITY: Trainee will be DCPO for a simple, random DC scenario consisting of single event (fire or flood), coordinating HQ1 and section base teams as the scenario develops to support the ship's DC organization and priorities. Trainee will be given changing DC priorities by (XO) assessor.

Personnel not being assessed as DCPO shall act as SBT I/Cs or Electrical manager in support of DCPO. DCPO Trainee may create and use an Aide-Memoire (eg. Duty Tech style index card).

Pass/fail criteria: Trainees to achieve all critical and 85% of checklist items to succeed.

ASSESSMENT RESOURCES:

- DC boards (to match BDCS trainer);
- BDCS trainer (DCTF preferred);
- Primary / Alternate comms (telephones, PRCs);
- DC-based scenarios; and
- Role-players (candidates not presently being assessed).

IMPACT OF FAILURE: Remedial Training.

PO2 MECHANCIAL SUPERVISOR – EC 003.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on 85% to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test

	ITEM	SAT	UNSAT	COMMENTS
	Initial Actions			
1.	*Establish communications			
2.	Identify Ship's status to include:			
	a. Fire Pumps and Pressure			
	b. Ventilation			
	c. Availability of Fitted Systems			
	d. *Identify location of all of the damage			
	e. Assess all of the Damage			
3.	Formulate an attack plan to include:			
	a. Assign an Attack Team			
	b. Assign boundaries (Fire/Flood)			
	c. Consider use of Fitted Systems (verbally)			
	d. Plot plan on BDCS			
4.	*D/C Org. Reported Closed up in under 6 min			
	Resource Utilization			
5.	Properly identify hazards using the Risk Folio			
6.	Proper use of the MT&S and BCDS to assess the ships' stability			
7.	Use of the DCC operator to include Fitted System Management			
8.	Utilize recommendations by SBT I/C's to make sound decisions			
9.	Use of Personnel to include:			
	a. SBT Personnel			
	b. Ships Company			
	Direct SBTs			
10	Confirm established boundaries (Fire/Flood)			
11	*Inform SBT of Hazards			
12	Attack Teams assigned and Attack Routes detailed (as required):			
	a. FSB b. ASB c. #3 SBT d. Manning Pool			
13	Control information on the BDCS			

	ITEM	SAT	UNSAT	COMMENTS
	Lead HQ1			
14	Gather Info/Formulated Plan/Executed Plan			
15	Maintain Composure			
16	Demonstrate confidence			
17	*Keep DCO informed			
18	*Maintain control of D/C Organization			
19	Adjust plan based on input from command			
20	Manage resources (i.e. HQ1 does not have endless supply of personnel and or equipment)			
21	*Assess ship's stability			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 003.02 Submit a Post-Incident Report

1. Performance. Submit a Post-Incident Report.

2. Conditions.

   a. Given:

      (1) Scenarios;

      (2) DWAN workstation;

      (3) TI Template;

      (4) Assistance; and

      (5) References;

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall submit a post-incident report IAW the Fleet Technical Authority (FTA) template by:

   a. (**Med**) Carrying out a preliminary assessment following an incident to include:

      (1) Assessing (battle) damage to vessel (IT0130);

      (2) Surveying battle damage (IT0810) as applicable;

      (3) Assessing (battle) damage to engineering systems (IT0120);

      (4) Investigating major equipment failures (AFT1520); and

      (5) Compiling information from alternate sources.

   b. (**Med**) Drafting a post-incident report by:

      (1) Identifying types of post-incident reports; and

      (2) Completing sections of post-incident reports.

4. Time Required.

   c. EO – 300 mins to deliver training;

   d. EC – 400 mins (total for all trainees); and

   e. PC – 700 min (total for all trainees).

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Carry out preliminary assessment following an incident			
1. Assess damage to vessel (IT0130). 2. Survey battle damage (IT0810) (if applicable). 3. Assess damage to engineering systems (IT0120). 4. Investigate major equipment failures (AFT1520). 5. Compile information from alternate sources: a. Logbooks. b. Witness statements. c. Captured data (EHM). d. Visual media.	A10 G-21 Annex A (TI Process flowchart) A111080-0 / 1080-1	**Method / Media:** Case Study, SBL **Trg Environment:** Classroom **Activity Details:** The class will be given two different scenarios that detail a major mechanical/electrical failure. Trainees will discuss and provide input on the follow-on actions they would take to gather all the information required based on personal experience and actual events. Instructor will ensure all TPts are covered off by their discussion. One scenario will be investigated by the entire class. The second scenario will be investigated in syndicates of 3. Scenarios to be developed based on existing TIs which will give a standardized solution. **Trg Resources:** Scenario/sample data	50 per scenario (100 total)
LESSON OBJECTIVE: Draft post-incident report			
1. Identify types of post-incident reports: a. Technical Investigation (TI). b. Board of Inquiry (BOI). c. Summary Investigation (SI). d. Spill report message. e. Halocarbon release message. 2. Describe sections of each type of post-incident reports.	A2 A10 G-21 Annex A (TI Process flowchart) A11 1080-0 / 1080-1	**Method / Media:** EDI, case study. **Trg Environment:** Classroom **Activity Details**: Instructor will explain different post-incident reports, covering off TPts. Trainees will draft a spill report and a halocarbon release message based on an actual event. The instructor will be available for questions and feedback. **Trg Resources:** • DWAN workstation; • DND Message Drafter; • Example post-incident reports; • Scenario(s); and • References.	200

7. Test Details. EO 003.02 will be tested in EC 003.02 IAW table below and Annex K.

EC 003.02: Submit Post-incident Report

EC Time: 200 mins research, 180 draft and submit, 20 min debrief

ASSESSMENT ACTIVITY: Trainees shall be given a scenario involving a single significant incident (fire, fuel spill, mechanical failure, etc.). In groups of two (preferably one Elec and one Mech if possible), trainees will write a post-incident technical investigation (TI) after compiling data from given sources. Upon completion, reports will be submitted for instructor feedback based on the EC Checklist.

Pass/fail criteria: Trainees must achieve a SAT on 60% to be successful with this assessment.

Scenario to include (as a minimum):

- Event;
- Cause;
- Witness statements; and
- TOR.

ASSESSMENT RESOURCES:

- Scenario;
- DWAN workstation;
- References; and
- TI Template.

IMPACT OF FAILURE: Remedial Training.

PO2 MECHANCIAL SUPERVISOR– EC 003.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

3. PASS / FAIL CRITERIA: Trainees must achieve a SAT on 60% to be successful with this assessment.
4. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
1. Introduction			
References are applicable to the incident			
All applicable references which were given are listed			
Scope of TI is clearly defined			
Scope of TI is in line with the TOR			
Aim of TI is clearly defined			
Aim of TI is in line with the TOR			
2. Background			
Relevant details that lead up to the event are captured			
References detailing lead-up events are properly included (eg. “Ref A”)			
Details of the event are captured			
References detailing the event are properly included (eg. “Ref A”)			
Details are paraphrased from the given sources (not word for word)			
3. Analysis			
Possible cause(s) are detailed			
Information from sources is paraphrased (not word for word)			
Logical thought progression is demonstrated through detailing of analysis process			
4. Discussion:			
Details of investigation are detailed			
Logical thought progression is demonstrated by progressive flow of discussion points			
5. Conclusion			
Clear and concise synopsis given of what happened			
Clear and concise synopsis given what was found			
6. Recommendation 1 (minimum of 2 recommendations)			
Relates directly to the incident (logical connection)			
Matches Commander’s Intent from TOR			
Meets SMART criteria (Specific, Measurable, Attainable, Realistic, Time-bound)			
7. Recommendation 2 (minimum of 2 recommendations)			

ITEM	SAT	UNSAT	COMMENTS
Relates directly to the incident (logical connection)			
Matches Commander's Intent from TOR			
Meets SMART criteria (Specific, Measurable, Attainable, Realistic, Time-bound)			
8. Format			
1000-2000 words			
Military letter format (address, heading, signature block, dist. list, etc.)			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 003: Manage Battle Control Organization

PC Time: 50 mins + 20 mins debrief per trainee (Total 700)

ASSESSMENT ACTIVITY: Trainee will be DCPO for a complex, random DC scenario consisting of multiple events (action stations, emergency stations, etc.), coordinating HQ1 and section base teams as the scenario develops to support the ship's DC organization and priorities. Trainee will be given changing DC priorities by (XO) assessor.

Personnel not being assessed shall act as SBT I/Cs or Snr ET in support of candidate being assessed. Trainee may create and use an Aide-Memoire (eg. Duty Tech style index card)

Pass/fail criteria: Trainees to achieve all critical and 85% of checklist items to succeed.

ASSESSMENT RESOURCES:

- DC boards (to match BDCS trainer);
- BDCS trainer (DCTF preferred);
- Primary / Alternate comms (telephones, PRCs);
- DC based scenarios; and
- Role-players (candidates not presently being assessed).

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt.

PO2 MECHANCIAL SUPERVISOR – PC 003 / EC 003.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with asterisk *) and achieve 85% SATs on the remainder of the checklist items to pass this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

	ITEM	SAT	UNSAT	COMMENTS
	Initial Actions			
1.	*Establish communications			
2.	Identify Ship's status to include:			
	a. Fire Pumps and Pressure			
	b. Ventilation			
	c. Availability of Fitted Systems			
	d. *Identify location of all of the damage			
	e. Assess all of the Damage			
3.	Formulate an attack plan to include:			
	a. Assign an Attack Team			
	b. Assign boundaries (Fire/Flood)			
	c. Consider use of Fitted Systems (verbally)			
	d. Plot plan on BDCS			
4.	*D/C Org. Reported Closed up in under 6 min			
	Resource Utilization			
5.	Properly identify hazards using the Risk Folio			
6.	Proper use of the MT&S and BCDS to assess the ships' stability			
7.	Use of the DCC operator to include Fitted System Management			
8.	Utilize recommendations by SBT I/C's to make sound decisions			
9.	Use of Personnel to include:			
	a. SBT Personnel			
	b. Ships Company			
	Direct SBTs			
10.	Confirm established boundaries (Fire/Flood)			
11.	*Inform SBT of Hazards			
12.	Attack Teams assigned and Attack Routes detailed (as required):			

	ITEM	SAT	UNSAT	COMMENTS
	a. FSB b. ASB c. #3 SBT d. Manning Pool			
13.	Control information on the BDCS			
	Lead HQ1			
14.	Gather Info/Formulated Plan/Executed Plan			
15.	Maintain Composure			
16.	Demonstrate confidence			
17.	*Keep DCO informed			
18.	*Maintain control of D/C Organization			
19.	Adjust plan based on input from command			
20.	Manage resources (i.e. HQ1 does not have endless supply of personnel and or equipment)			
21.	*Assess ship's stability			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

ANNEX A - *MAIN REFERENCES*

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	C-23-HFX-000/MS-003 Manual of Trim and Stability
A2	Halifax Class Safety Environment Management Systems (SEMS)
A3	C-03-005-033/AA-00 Naval Engineering Manual (NEMS)
A4	C -82-007-001/AF-000 Reference book on petroleum and associated Products (Petroleum oil and lubricants (POL) handbook)
A5	C-82-020-000/TB-001 Technical Guidance for the Procurements of Fuels and Lubricants by Deployed Ships
A6	Ships Standing Orders AL11
A7	Canadian Forces Control Documents 132
A8	Risk Folio C-97-007-000/DC-003
A9	STG (Sea Training Group) http://halifax.mil.ca/SEA_TRG/
A10	MARPACORD 1775-6 HMC Ships and HMC Dockyard ESQ
A11	NAVORD 1080-0 Technical Investigations – Policy
A12	MARPACORD 19000 series MARPAC FUEL MANAGEMENT
A13	NAVORD 3001 In-Service Naval Material Risk Management Policy and Process
A14	NAVORD 3250-7 Material Operational Deficiency Process
A15	NAVORD 11200 Readiness State Transfer

ANNEX B - *TASK LIST*

Task	Description	T/NT	NT Reason	PO
AFT1510	Inspect completed repair or installation work	NT	Move to PO2 Common	
AKT0060	Analyze bearing parameters	NT	Gained through experience	
AMT0030	Analyze coating quality	NT	Trained at LS Mech Spec USQ	
AMT0460	Inspect welding procedures and process	NT	Trained on Welder USQ	
AMT0760	Survey coatings	NT	Trained at LS Mech Spec USQ	
AMT0880	Inspect all metal welds and parent metal for defects (Destructive)	NT	Trained on Welder USQ	
AMT0890	Inspect all metal welds and parent metal for defects (Non-destructive)	NT	Trained on Welder USQ	
AOT0930	Manage fresh water system	NT	Trained at AB USQ	
AT0300	Trial ship's Citadel (gas tight)	NT	Gained through experience	
AVT0180	Analyze diesel engine cylinder firing pressures	NT	Trained in LS MECH, AIBX and DI course	
BAT0120	Inspect gear mesh	NT	performed by gear inspector	
BT0080	Inspect main propulsion gearing	NT		
BT0090	Inspect main propulsion shafting	NT		
BT0100	Inspect ship's (not a small boat) propeller	NT	Gained through experience	
GT0020	Advise on undocking/docking preparations	NT	Gained through experience	
IT0050	Advise command on hazardous material (HAZMAT) organization team actions (MCDV)	NT	Gained through experience	
IT0450	Manage hazardous material dressing area	NT	Trained on PO2 Common	
OT0010	Advise command on fuel quality	T		002

OT0020	Advise on maritime vessel fuelling operations and readiness	NT	From RQ-PO1	
OT0080	Lead fuel spill response team	NT	Gained through experience	
OT0090	Lead maritime vessel fuelling (internal fuel / providing / consolidation)	T		002
OT0110	Manage fuel quality testing	NT	Gained through experience	
OT0210	Supervise liquid cargo transfer (SEMS)	NT	Gained through experience	
UT0160	Manage confined space rescue	NT	FLYCO	
UT0300	Supervise confined space rescue	NT	FLYCO	
WT0010	Adjust ship's stability	T		001
WT0030	Analyze ship's stability data	T		001
WT0040	Assess ship's stability	T		001
WT0050	Investigate ship's stability	T		001
WT0080	Monitor ship's Draught	T		001
WT0090	Monitor ship's stability	T		001
WT0110	Record ships stability variables	T		001
WT0120	Trial ship's stability	T		001
ZT0090	Supervise docking preparations	NT	Gained through experience	
ZT0110	Supervise undocking preparations	NT	Gained through experience	
*IT0350	Manage battle damage control organization	T		003
*IT0130	Assess damage to vessel	T		003
*IT0810	Survey battle damage	T		003
*AFT1520	Investigate major equipment failures	T		003
* IT0120	Assess damage to engineering systems	T		003

Note: Tasks marked with * are originally from Mar Tech Mgr RQ PO1 and have been added as part of PO 003 Manage DCHQ.

ANNEX C - *QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS*

QSP Finalization and Review Notes and Recommendations:

For next review/board:

Areas such as stability and DC roles/responsibilities should be coordinated across RQ PO2, USQs, and RQ PO1 so they are tied together, build on previous tasks, and avoid and gaps in training (and tasks).

Fueling I/C: Recommend this PO have a deeper review and include an OJPR component

Fueling I/C tasks need an OJPR – this PO "qualifies" someone as fueling I/C.
These tasks and training are not part of SWK package but was part of the old cert package – now graduates are "qualified" fueling I/C but only have ever done it electronically.
Sems SOP EO1 Part 2, para 8b still states a Cert 3 or above is required for this position.
A TM that states SWK means Cert 3 – where is this – if true this must be included in SWK training as well.

Dates: 16 Apr 18 – 27 Apr 18

Location: Workpoint TB 111

Board Members:

Rank and name	MOSID / Unit	Dates
CPO2 J. Zsombor, Board Chair	00379-06	16 – 27 Apr 18
CPO2 J. Drew, SME	00379-06	16 – 27 Apr 18
PO1 G. Buckle, SME	00379-06	16 – 27 Apr 18
PO1 C. Switzer, SME	00379-01	16 – 27 Apr 18
PO1 S. Mann, SME	00379-01	24 – 27 Apr 18
Lt Tzotzos, TDO	00211	16 – 27 Apr 18
A/SLt Kyle Knight, Secretary / TDO OJT	00207 / NTDC(P)	16 – 27 Apr 18

Board Objective:

To analyze the job performance and design the training requirements for Marine Technician PO2 Mechanical Supervisors based on the JTAR and legacy requirements of the amalgamated technician trades.

Discussions / Decisions:

***Day 1 16 Apr 18**	Reviewed Mar Tech Supervisor Job	Board came up with what they though the job responsibilities would include.
	Reviewed Mech Supervisor/Electrical Supervisor primary responsibilities from JBS	Found that many of the responsibilities identified by the board were there, signifying the board had a good understanding of the job. The responsibilities were similar between the Mech/ Elect generating much discussion if there should be two separate courses or push both to the common course. Group felt the discussion was premature and as the process went on this would be better defined.
	Stability	Board had much discussion on the 8 stability tasks. There is 3 days of training on the PO1 common course. The concern is that without training at the PO2 level the Elec would be at a disadvantage of the Mech, with advanced training. As well the Mech would receive duplicate training at the PO1 level. The board brought in the PO2 Electrical board to discuss. The general consensus was that the training at the PO1 level would be adequate and that they would have little trouble with the concepts. They were concerned that the training would have to change if the Elect were to have to perform as DCPO but that is currently not anticipated. The two boards voted on three COAs, bulk of training, 2-3 weeks, at the (1) PO2 Mech, (2) PO2 Common, and (3) PO1 Common. The boards chose to leave at the PO2 Mech.
	Task list	Board reviewed task list and answered questions found in the comments column.
	Missing tasks	Board reviewed the NT train list to see if any tasks should be moved into the task list. No tasks were moved.
	Trials	Board defined trial that were discussed on the common RQ but may need to be specialized in the Mech RQ. Tasks were left to be scalared to determine actual gap in performance from the P2 RQ.

	Conducting Trials	Board scalared the existing tasks. Original Task were Trial Ships Stability and Trial Ships Citadel. Added were MMPT, Steering, Pitch, and lifting appliances. TDO identified many of the tasks identified under the existing tasks are actually teaching points. The board feels that the teaching points are valid, however it was difficult to identify the performance and a PC. Looking into the JTAR, machinery trial was a NT at the PO1. The board suggested that this was an experience piece. Task, Trial Ships Stability would move to common. ***Update: This task was included to the incline experiment in PO 001on the PO2 MECH Supr.***
	Fueling	OT0090 Lead Maritime Vessel Fueling caused much discussion as this is not a task on its own and it incorporates alongside and at sea replenishment, defueling, transferring and small craft. Even though this task had been discussed at the MS board several tasks had to be re-defined. The outcome of discussions from the beginning stages of the now defunct MS Sr Mech Tech QSPWB, was that D Nav P&T directed that fuelling is a PO2 Mech Supr duty and will be trained at this level. Much discussion was placed around if the training could be provided in the school or if OJPR or EQ were required. This will be determined once EOs are developed.
	Stability	The board discussed if the legacy training from the HT 6A can be applied directly into the Mar Tech Sup. That training is approximately 15 days. The PO1 training is 1200 minutes. The board feels if this training was meant to be at the PO1 level, it was underestimated how long it would be to teach. The board plans to vet the legacy training to ensure that it fits the tasks and expected performance of the PO2 Mech Supr.
Day 2 17 Apr 18	Stability	The board discussed the stability PO 002 from the former HT QSPs IOT try to discern the amount of stability and information that needs to be on this new board. The board informed the TDO that this was a very crucial point and that they should get input from the TM to ensure they are covering off the pertinent points.
		The board discussed that there will be 3 teaching points of calculations that will be a carry on in the PO1 from the information they learnt in the PO2. The board needs to focus on not duplicate training and they will expand on this when they develop the teaching points.

	INELP (Personnel management)	The TDO from the Elec board came to discuss when we should teach the Watch and station bill information. They discovered that it was not taught at any point on the RQPO2 or the Mech/Elec Supr courses. After discussion the board decided that this information should be taught on the RQ PO2, however how to deliver this information is difficult because this information may be all experience based. There was talk to assess the INELP program because that would be a good location to include it. The board then began examine the location of the INELP training and the Elect TDO stated that they would consult their SMEs.
	AOT0930 - Manage fresh water system	The board discussed if this was a task for the PO2 and if it was trainable, this job may be an experience piece and that would make it a NT. The board reviewed the OS, AB and LS QSP and determined that this was taught in the AB, however it did not adequately cover the requirements for collecting water samples.
		The board identified that the training about taking proper water samples needed to be improved or "beefed up" at the AB level (EO 009.01). Follow on action after this board is to look at adding more training to the AB IOT ensure their confidence and proficiency when it comes to properly taking/collecting water samples.
	ZT0090 – supervise docking preparations GT0020 – Advise on undocking/docking preparations ZT0110 – Supervise undocking preparations	The board discussed the Mech Supr's job during docking/undocking which was to supervise the conduct of the docking plan which is created and managed by the Mech Mgr. Because it is only supervising the tasks performed by subordinates, there is no new knowledge or skills required to supervise these tasks. Outcome of discussion is that all tasks related to docking/undocking are NT: - GT0020 – Advise on undocking/docking preparations - ZT0110 – Supervise undocking preparations - ZT0090 – Supervise docking preparations
	AOT0930 – Manage fresh water systems	The board decided that it was a NT, this aspect is experience based and no new knowledge or skill is required to be trained to manage the system.
	AVT0180 Analyze diesel engine cylinder firing pressures	The board decided that this task was a NT because it is covered in LS Mech, AIBX and DL Course.
	AT0300 – Trial ship's Citadel (gas tight)	The board determined this was a NT because there is nothing new besides trial procedures which is trained on RQ PO2 course.

	Survey of Welding and Coatings	Tasks related to surveying coatings and welding are NT. It is recommended that tasks related to coatings be moved to the LS Mech PO 015 on polymers where trainees learn to apply and repair coatings. The actual TPts already exist so the task numbers would only need to be added. No extra trg time would be required: AMT0760 – Survey coatings AMT0030 – Analyze coating quality It is recommended tasks to inspect metal welds through destructive and non-destructive means be moved to the LS USQ welder course because it will be the Mech Supr with this USQ qualification with combined experience who will be completing these inspections. Rather than train all Mech Suprs who may not have the welding expertise, it should be covered at the level where the Mech Tech is trained to weld so that they have the opportunity to develop these skills as they progress in rank: AMT0460 – Inspect welding procedures and process AMT0880 – Inspect all metal welds and parent metal for defects (DT) AMT0890 – Inspect all metal welds and parent metal for defects (NDT)
18 Apr 18 Day 3	AFT1510 – Inspect completed repair or installation work	NT. This has been gained through experience as Sr LS and MS that it is not trained at the Mech Supr level.
	Stability	The board will meet with the Elec Supr QSPWB to further discuss the possibility of moving this section to the RQ PO2 common. While Elec board stated earlier in the week that their current level of math will allow them to perform stability calculations on the PO1 Mech Mgr course, the full scalar of this PO has determined that there may be a larger gap in performance between PO2 and PO1 which begs a second review.
	Target Population Analysis	The board completed the population analysis and determined: • qualified technicians with a large breadth of experience to rely on for this course • 8 – 10 years' service • hands on learners who would respond best to problem and scenario based instruction • comfortable with group learning methods

		• Training is located on both coasts and has the potential for people with potential for trainees to proceed on TD from other locations • The trade consists of both genders, however largely male • minimum of grade 10 education and some with college tech courses • environment: hard sea trade • Age range: early 20s onward • interested in how things work (mechanically inclined) and driven problem solvers • many have families and interests outside their working hours • The members have 8-10 years in on average
	Tiered Readiness Discussion with Elec Supr Board	PO2 Mechanical board was invited to the Elec Supr board to discuss the merits of the inclusion of portions of the QL 6b Electrical legacy QSP. Specifically, PO 002 TP f.5 related to tiered readiness programs and PO 003 departmental maintenance were discussed. It was unanimously decided that the content of PO 003, once updated to account for recent changes in management programs was to be incorporated in the PO2 RQ course. The content of EO 002.02 F.5 was determined to be performed at the PO1 RQ level.
	RQ PO2: Increase Stability and include personnel management	The PO2 Mech and PO2 Elec board members agreed on including stability and personnel management content (aforementioned) in the PO2 Common course. The Mech and Elec portions will be separate USQs immediately following the PO2 RQ Common. Further discussion with MS Div Comd, TM and D Nav P&T is required to make such changes.

19 Apr 18 Day 4	D Nav P&T meeting information	Elec/Mech P2 Board chairs/TDOs, RQ P2 TDO, MS Div Comd, TM and D Nav P&T had a telecon to discuss proposed changes to P2 USQs (mech/elec) the RQ PO2 and RQ PO1. From mechanical perspective, requested changes included: - Increasing stability at PO2 RQ to give all Mar Tech Suprs a stronger foundation for them to develop their abilities to to be capable at PO1. - The division would be emergency/damage stability at PO1 and remainder of stability calculations and theory would be moved to RQ PO2. Direction from D Nav P&T: - Stability as it is laid out in the RQ PO2 would remain and most stability from PO1 would be removed and pushed down to the PO2 Mech Supr. There was never any intent for stability to be framed from an emergency/damage perspective at the PO1 level so it is not required there. - **DCPO is to be removed from PO1 Mgr and moved to PO2 Mech Supr. This is because Elec Mgrs were never intended to be DCPOs. As a result, this trg will be moved to PO2 Mech Supr even though they will not act as DCPO until PO1. PO2 Mech Supr TDO suggested a module of "just in time" trg that could be created and accessed by those ready to fill this position when it is required. D Nav P&T didn't support this**: ***Update from CPO1 Pryor see 25 Apr 2018***
		A member of the board believes that giving the trainee Section Base IC at RQ PO2 and then DCPO on PO2 Mech Supr will cause a critical manning issue. Training prior to them acting as DCPO at the PO1 rank could result in skill fade and diminishes a tiered approach to training. They are putting all the training on one level. Members of the board worried that this new course of action may deprive the new trainees of the ability to get valuable on the job experience required for DCPO.
		The board wants to remove parts of PO 004 from RQ PO1, including EO 004.02 and 004.03 to create a new DCPO for the PO2 Mech Supr course. Those unique electrical tasks not required as part of the DCPO will be removed.
		The board decided to move the entirety of PO 002 – Manage Ship's Stability, from RQ PO1to PO2 Mech Supr course.
	Definition: operational ship	A ship at sea or along-side, all systems equipped and operational.
	Fueling IC	The board decided that to properly teach the fuelling IC PO it was necessary to group all of the paperwork and administration in the first EO because it will give the trainee a better overview of the process and inform them of the step by step administration that's

		required for this evolution. As a result, EOs appear different from the PO standard. Nonetheless, the standard is still adhered to even with this shuffle to improve training delivery.
19 Apr 18 Day 5	Fuelling IC – discussion of whether this is gained through experience onboard ship or if there is a gap in performance that can be trained.	The board decided that scenario-based training was the best way to communicate the information to the students about analyzing the fuel samples.
		After discussion the board discovered that the IC PO may be more of an experiential piece which would mean that some aspects of this should not be trained.
		Some members of the board believe that without the redundant training members will be lacking that due diligence knowledge cause a problem. The TDO advised the board that gaining confidence of a ship's crew and due diligence are acquired through experience, practice and time. This aspect can't be trained.
		Another member believed that because of the different experience levels Mech Suprs may have relating to fuelling evolutions, fuelling requires training for the foreseeable future.
		The board chair outlined 4 COAs to proceed with Fuelling IC: 1. Leave the training as they have it (looking like legacy) 2. Add a delta (which is a no go) 3. train it under the new construct to standardize the fuelling process; and 4. EQ
		A member of the board then put forth the idea that this could go to the Sr. watch keeper level, however since Sr WK is not mandatory, a cross-section of Mech Suprs would not be trained to act as Fuelling ICs.
		The board chair decided that they would build something that looks like the Legacy, that has very little new training and understanding those limitations he believes that they still need this training. If this turns out to be a mistake this will become clear at delivery.
		The Chair also believes that they cannot just NT it because there is no capability, they need this information to be reinforced from the LS lvl which is an amalgamation of all the skills they learned from the LS.
		The board voted and the results were as follows - PO1 Switzer abstained - PO1 Buckle voted to continue - CPO2 Zsombor voted to continue The decision was made to continue developing a PO similar to what currently exists in legacy.
		LCdr Day, MS Div Comd, then came to speak with the board and after hearing the boards suggestions, LCdr Day said that after looking at this it is well suited to an EQ because it is most likely a consolidation of overall experience to perform this duty.

		The TDO suggested “Just in Time” training where anyone could access the module through DLN on Fuelling IC at any time. The Div Comd suggested that it may be too difficult to manage if there was no qual/MITE attached to it.
	Stability	The TDO from the RQPO2 and RQ PO1 and the MS Div Comd also discussed stability and how much should be included on the PO2 and PO1 course. It was determined that stability on the PO2 Mech Supr should be critical in determining the level to which stability calculations must be made. The board will pay close attention to this when developing the PO.
		To bridge the performance between PO2 Mech Supr and RQ PO1, since DCPO has been pushed to the Mech Supr course, the Div Comd suggested stability calculations be removed from the RQ PO1 as the PO1 would just need basic knowledge and understanding of the terminology as they would use GLM (or similar) to confirm calculations. Likewise, the PO1 Mgr would rely on the PO2 Mech Supr to provide SME support on stability calculations.
		The Div Comd still believes that calculations are required but it should be taught to a less advanced lvl because it is just not used enough to make it worthwhile.
	Fuelling IC	The board designed a series of COAs for the role of fuel IC in order of recommendation: - Short train and test as OJPR with a short administrative review in class with a PC completed on ship, this should be made class specific ITO address any issues that come up - Move the fuelling IC to the PO2 common course, they see no reason why the elect cannot be trained as a fuelling IC furthermore it increase the utility of the members. - EQ - NT (the board does not like this as a COA at all)
		The board decided that they would continue to develop the PO taking into consideration previous experience. PC assessment will be developed as an OJPR onboard ship post course completion to confirm performance standard has been achieved. An onboard PC is seen as necessary because SMEs indicate it can’t be adequately assessed in the TE.

	Main Propulsion Components becomes NT BT0080 – Inspect the main propulsion gearing, AKT0060 – Analyze bearing parameters, BT0100 – Inspect ship's (not a small boat) propeller, BAT1020 – Inspect gear mesh	The board reviewed the following references related to gearing and main propulsion component inspections:C-24-302-000/MP-001 Main Propulsion Gearing Manual Volume 1 of 2 Text, Tables and Lists and C-03-005-033/AA-000 Naval engineering manual (NEM) These references identify that gearing inspections are not performed by the PO2 Mech Supr but rather the gearing inspector from RF. The PO2 Mech Supr's job is to supervise closure of gearbox after inspection. Review of the LS Mech Tech identified that gearing theory and overview is extensive. As a result, the PO2 Mech Supr will have gained sufficient experience working with gears in various capacities during this time, that there is no further knowledge and skills trainable at this level to support this performance. The board originally intended to build a PO IAW those tasks however they determined that it was a NT due to EXP, and the fact that the part of the job was no a PO2 task according to Refs. This was the cut that left this course with 3 POs to which are Stability, Fuelling I/C and DCPO
23 Apr 18 Day 6	Completion of Fuelling IC PO	The TDO went over the information and made some recommendations about the fuelling IC PO. The board finished the fuelling IC PO. The board was asked again if trainees should have the opportunity to practice acting as the fuelling IC prior to their EC. It was stated that the presentation activities trainees will do for EO 002.02 combined with their experience being part of fuelling parties, that there was no requirement to have them run through an evolution prior to the EC.
		The board discussed EC 001.01 and if it was necessary to give them all of the fuelling check lists. Some of the lists consist of information that is not applicable to actually test in the administrative aspect of the order so the board decided that this was unnecessary to test. The board's formal decision was that they are going back to the written assessment using scenarios because it is the best way to assess their ability to work with the checklists.
		The board decided that this PO will be brief because all of the members on this course will already have a breath of experience covering various pieces of fuelling. The board decided that there was no need to belabour this information through extensive practice in the delivery of training.
		The board decided that there was no need to train information concerning fuel spills because the member would have already received the training in the Duty TECH course. (Task OT0080 Lead fuel spill response team)

24 Apr 18 Day 7	Review of the RQ PO1 PO002 Submit Post-Incident Report	The board discussed if it was necessary to have the member write a TI for an EC and a PO also on a TI. The board believes that the Elec missed the mark for what they believe is necessary for the Mech to know.
		The board discussed if the students require time in class as well as time at home. The board chair decided that this should be assigned and given time in class to work on it allowing time at home if required.
		A board member bough up the point that because this has been looked at twice already are we just wasting our time attempting to reinvent the wheel once more. The EO remains as it was written for the RQ PO1 course.
	CTAT References	All references written in red are CTAT and have been highlighted in red to signal as such in the event development is contracted to civilian companies.
	PO1 Mann	PO1 Mann joined the board to support design of stability and DCPO POs because of his experience.
	Stability PO 001.01	The board decided that the task WT0120 which they had decided was a NT fit well into this PO and decided to train it in 001.01 because it fit well with the requirement to perform "incline tests"
		The board discussed if what they had completed in the stability PO was missing the mark because it was a very linear math oriented process. The board began looking at ways to inject performance into these events rather than just punching numbers into a calculator.
		The board talked about moving the entire EO 2 from the PO1 course into the PO2 course. An issue identified with this is that the PO1 use the "management" term that was originally not applied to the PO2. The board decided that this EO will be moved in its entirety into the PO1.
		The board learned that stability videos used in NWO training could be utilized to support delivery of training on the stability PO.
	EO 001.01	The board decided that this EO did not need to have an EC because this information is the background that will be used for all other calculations and information moving forward. This information will be tested on a later EC.
25 April 18 Day 8	PO 002 Stability	The board discussed what the correct level of training should be, finally making the decision to maintain a level of training equivalent of the legacy training.
		Direction from D NAV P&T was received from CPO1 Pryor (OccMgr) the forwarded e-mail follows: "From the PO1 RQ – PO 002 is to be moved to the Mech Supervisor USQ this will include those Stability topics in Para two below. The intent here is to prepare the Mechanical supervisor for

		the Role of DCPO (this is not necessarily only on the "heavies" as the role may be at the PO2 Level on some classes). EO 004.01- Act as Electrical Manager for Emergency/Action Stations is to be moved to the Elect Supervisor USQ. EO 004.02 – Act As DCPO is to be moved to the Mech Supervisor USQ. EO 004.03 will remain at the PO1 level. I understand That within the PO1 RQ "Manage the DCHQ" is then split apart into the Mech and Elect functions but the Management of the DCHQ is not simply those functions it is the oversight and overall control of the DC Org. This responsibility lies with the DC Officer as advised by the DCPO (typically the MSEO on CPFs). The PO 004 "Manage the DCHQ" conceptually, must stay at the PO1 RQ without EO 004.01 and 004.02 but will have to focus on the role of the DCO (remember that MCDVs don't have NTOs – the PO1 is DCO). PO 003 (Docking and Undocking) will stay in the PO1 RQ in its entirety. Regarding the stability required for the PO 003: We are not required to do a fulsome re-teaching of all things stability. That which is required for docking and undocking may be more aptly termed "static stability lite" and needs to be focused on the evolution at hand. Everyone in the Occupation will have a generic understanding of basic terms and their meanings because all will (future) have watch keeping and shipboard experience as Mar Techs. What this means is that for docking and undocking the teaching of the process will have to include a review of the basic stability terms related to the process. Many will say that this "review" will be new knowledge for the Elect. Frankly, this for the Mar Tech is high school level trigonometry, so I don't think they will be as challenged as we assume. For clarification: this does not require any review of damaged stability terms or calculations. These are the domain of the PO2 Mech Supervisor." The board discussed the impact of the direction, and decided to continue working toward the PO2 as stated, recognizing further review of the PO1 training will be required. Much discussion was generated on the teaching delivery of the course. Traditional, instructor led work has been successful in the past and is a known entity. TEL options are available. Training similar to Khan Academy was discussed, with mixed experiences

		within the board. Training videos, currently available at the school are accurate, but possibly dated. Some members thought that the videos are actually so old that they have a novelty factor. A quick review of available training courses online identified a surplus of options. Assigning timings was challenging, as the board members felt they had a firm understanding of the complexity of teaching the tasks. Referring to legacy training found the times allotted did not immediately correlate to the tasks. The timings are likely to change as content is developed. **Stability Extra Practice Handbook: A stability handbook may need to be generated to provide opportunity to re-enforce concepts learned through the class and provide a take-away the student can use a future reference.** Created enabling objectives PO001 Stability. The board looked at duplication between P001 and P003 to identify where the grounding information should be.
		The board discussed if they should make the EC of 001.07 on their own, or if they should test this information about icing and grounding. They also questioned if they should EC the information at all or if it was an experience aspect. The TDO then suggested that this may be a good place to use a group activity or in pairs IOT be able to help the student step by step. The board decided that the EC should be on icing and grounding and the activity will be a complete walk thought of the entire process
		Delivery of EO 001.01 Identify Stability States and Forces Acting on a Ship, is a continuation of the introduction to stability presented in the RQ PO2, EO 002.01 Make Stability Recommendations to HQ1. The RQ PO2 course provides a general introduction to stability given that Electrical Supervisors need a cursory understanding. Stability, as part of the Mechanical Supervisor course goes beyond terminology and demonstrations only and focuses on the calculations, applications and relationships between the states of stability.

	DCPO	The board discussed how they will break up the DCPO EO to incorporate it into the PO2. There was discussion about if the students will need the more backing knowledge. A member of the board brought up the fact that these trainees may not dealt with sending multiple teams into a fire space. His thought was that they do not have enough training leading up to the actual event, and the overarching idea that they need more information before they can effectively handle an event. The board countered that the trainees are likely to have experienced these events The member countered with the idea that not all member may have had the same exp piece so he still believes that there needs to be more training to ensure that the trainee would be safe and make effective decisions.
		The board then discussed the possibility of adding an EO that pertained to building battle problems and scenarios. However after looking at the QSP they noticed that some of this information is addressed in different ways
	Decision DCPO	The TDO called for a decision, do they change it or do they leave it the same, the board decided to add some content on firefighting and formulating a risk assessment for moving teams and members into spaces with fire considerations. The added time to the Lesson objecting Direct SBT I/Cs to allow the students to talk about more types of fires.
	task Record ship's stability variables (WT0110)	The board decided that task Record ship's stability variables (WT0110) was a NT
3 May 18	D Nav P&T	CPO1 Pryor sent an email directing further direction on the QSP. Relevant excerpt below (date 3 May 18): So, I looked at the JTAR and the various notes and gave everything some deep consideration here is the direction regarding the PO1 RQ: PO 002 Will Remain in the PO1 RQ PO 003 (Docking and Undocking) will Remain in the PO1 RQ in its entirety. PO 004 EO 004.01-Act as Electrical Manager for Emergency/Action Stations is to be moved to the Elect Supervisor USQ. PO 004 EO 004.02 - Act As DCPO is to be moved to the Mech Supervisor USQ. PO 004 EO 004.03 will be common to both of the PO2 USQs. This doesn't mean that it is to be moved to the PO2 RQ but needs to be present in both the Mech and Elect Supervisor USQs. Note The Role of the DCPO/ DCO on the MCDVs will be covered off by a separate piece of training (TBD if required). This is different from previous direction from CPO1 Pryor email dated 25 Apr 18:

		From the PO1 RQ – PO 002 is to be moved to the Mech Supervisor USQ this will include those Stability topics in Para two below. The intent here is to prepare the Mechanical supervisor for the Role of DCPO (this is not necessarily only on the "heavies" as the role may be at the PO2 Level on some classes). EO 004.01- Act as Electrical Manager for Emergency/Action Stations is to be moved to the Elect Supervisor USQ. EO 004.02 – Act As DCPO is to be moved to the Mech Supervisor USQ. EO 004.03 will remain at the PO1 level. I understand That within the PO1 RQ "Manage the DCHQ" is then split apart into the Mech and Elect functions but the Management of the DCHQ is not simply those functions it is the oversight and overall control of the DC Org. This responsibility lies with the DC Officer as advised by the DCPO (typically the MSEO on CPFs). The PO 004 "Manage the DCHQ" conceptually, must stay at the PO1 RQ without EO 004.01 and 004.02 but will have to focus on the role of the DCO (remember that MCDVs don't have NTOs – the PO1 is DCO). **End state:** PO 002 will remain at RQPO1. EO 004.02 Act As DCPO was already moved. EO 004.03 Submit a Post Incident Report will be moved back down to the RQ MechSupr as originally intended by the board.
	Addition of Task to PO2 MechSupr From RQ PO1	To support Act as DCPO, the following tasks will need to be moved from RQPO1 task list to the USQ PO2 MechSupr: (1) Assess (battle) damage to the hull (IT0130); (2) Survey battle damage (IT0810) as applicable; (3) Assess (battle) damage to engineering systems (IT0120); (4) Investigating major equipment failures (AFT1520); and (5) Manage battle damage control organization (IT0350),
	Change of PC 002 Assessment	Board had changed PC 002 from being performed onboard ship upon completion of residential course to being assessed in the schoolhouse. Original discussions didn't consider how trainees with shore postings would be able to complete the PC without having a ship to do so. The impact would be trainees may be waiting months to do the PC before they receive the qualification. This is not prudent and lengthy delays may result in qual never being granted. Discussions are still being made that an EQ would serve to bridge the gap between training in the schoolhouse and providing the Mech Supr sufficient practice and experience to act as fuelling I/C in full capacity. This detail remains outstanding and delivery of pilot course and follow on evaluation/validation will verify this.

Lessons Learned:

1) The board suggested that a step to add to the board process would be to consult DNav P&T once the tasks have been grouped IOT ensure that the board is meeting the

requirements that they require to avoid costly delays in board processes when requirements shifted or weren't clearly defined at the outset.

2) The board believed that the format of this QSP had pros and cons. One advantage is that developers (CCD analysts or contractors) will have clear direction in how lessons are to be developed. The downside is that due to the extremely short timeline of the QSPWB, full consideration of teaching points and instructional methods can't always be made. Often during lesson development, more in depth deliberations results in well defined instruction. This format doesn't allow for the deliberation.

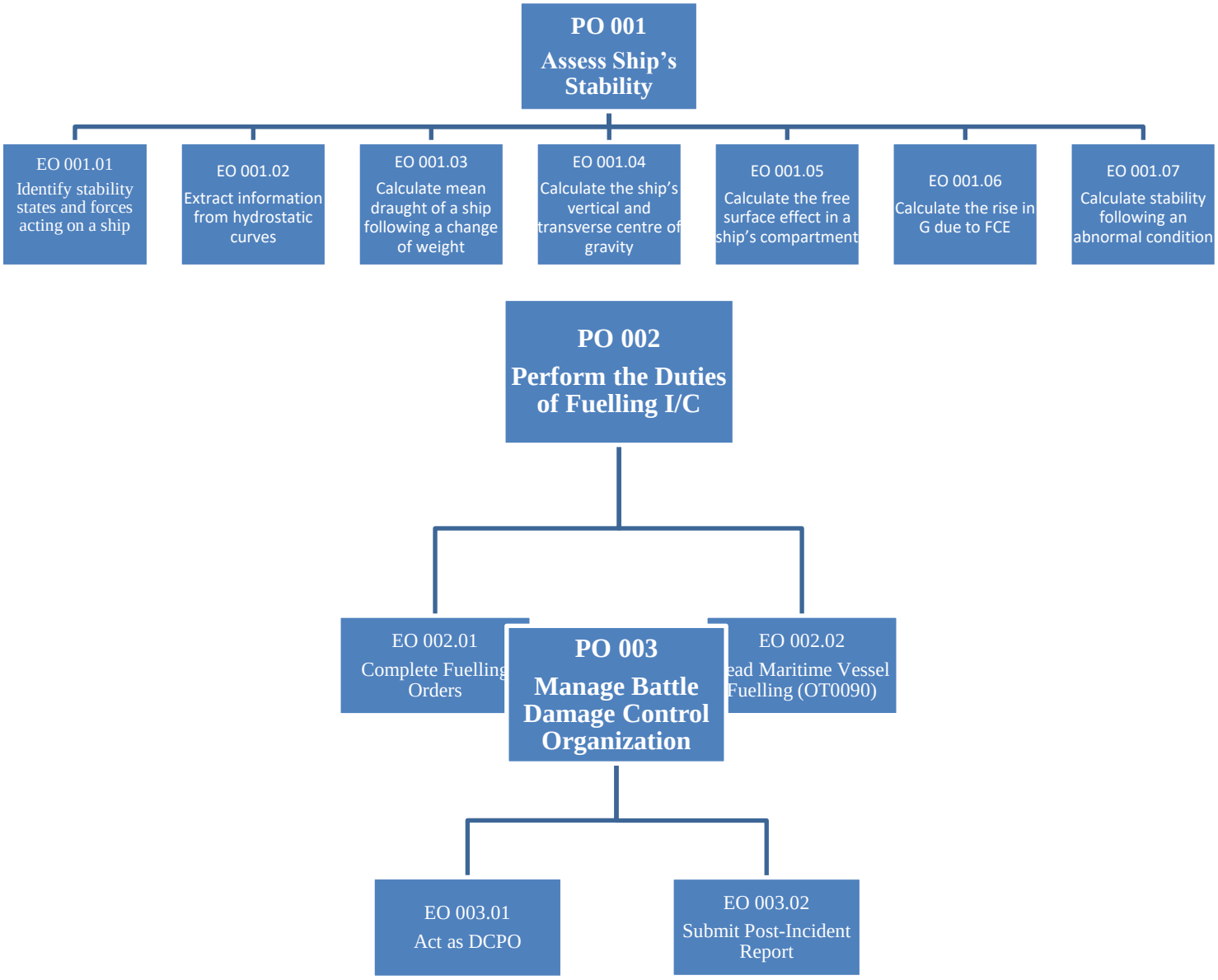
Guidance to Instructional Staff: It is important to review this QSP prior to instructing the course. It contains important details related not only to what to instruct, but timings and the necessary resources to deliver the course.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Add tasks from Manage DCHQ tasks from RQ PO1 to JTAR. • Assess (battle) damage to the hull (IT0130); • Survey battle damage (IT0810) as applicable; • Assess (battle) damage to engineering systems (IT0120); • Investigating major equipment failures (AFT1520); and • Manage battle damage control organization (IT0350)	TDO	
2	Review changes from RQ PO2 and RQ PO1 to ensure all QSP changes accurately captured and actioned in advance of mar Tech Alignment	TDO	
3	Tasks be added to PO2 MechSupr on MTL	D Nav P&T	

Other Considerations: Nil

ANNEX D - *QSPWB SCALAR*



MARPAC RDIMS #
Chap 1 approved DD MM YYYY
Chap 2, 3approved DD MM YYYY

ANNEX E - *ON JOB PERFORMANCE RECORD (OJPR)*

No OJPR required. To be updated if status changes.

ANNEX F - *ACRONYMS AND ABBREVIATIONS*

Acronym / Abbreviation	Term or Definition
AMD	Draft at midship
B	Centre of buoyancy
BDCS	Battle Damage Control System
BOI	Board Of Inquiry
DC	Damage Control
DCC	Damage Control Console
DCPO	Damage Control Petty Officer
DCTF	Damage Control Training Facility
DFTK	Diesel Fuel Test Kit
FCE	Free Communication Effect
FSE	Free Surface Effect
FTA	Fleet Technical Authority
G	Centre of Gravity
GLM	General Load Monitor
GM	Metacentric height (vertical distance between G and M
GZ	Righting Arm Lever
IA's	Initial Actions
I/C	In Charge
IPMS	Integrated Platform Management System
KB	vertical distance from B to Metacentre
KG	Vertical Distance From Keel to VCG
KM	Height of Metacentre (vertical distance from keel to Metacentre)
LBD	Length Between Draughts
LCF	Longitude Center of Floatation
LRH	Leadership Respect Honour
LSO	Landing Safety officer
PCOT	Propulsion Control Operator Trainer
PERS	Personnel Evaluation Reporting System
PM	Preventative Maintenance
POL	Petroleum Oil and Lubricants
PPE	Personal Protective Equipment
RAS	Replenishment At Sea

RAT	Rapid Attack Team
RF	Repair Facility
RIT	Rapid Intervention Team
ROE	Record Of Events
SBT	Section Base Team
SCBA	Self Contained Breathing Apparatus
SOP	Standard Operating Procedure
SSO	Ships Standing Orders
SWP	Short Work Period
SWOAD	Ship With Out Air Department
TBD	Trim Between Draughts
TI	Technical Investigation
UCR	Unsatisfactory Condition Report
USQ	Unique Specification Qualification
VCB or KB	Vertical Centre of Buoyancy
VCG or KG	Vertical Centre of Gravity

ANNEX G - *SUGGESTED SEQUENCING AND TIMINGS*

1. This annex is to be used by Campus Support to properly schedule all required classrooms, trainers, ships, personnel, etc for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
2. The tables are completed in the preferred sequence for lessons and resources. They are to be read in conjunction with the proposed course schedule (when available).
3. For modularized courses, the tables shall be divided based on module.
4. Special instructions to schedulers shall be included as “Notes” following the applicable PO or Module and can include:
   a. Elements which must be scheduled in a particular order;
   b. Elements which can be scheduled in any order;
   c. Common scheduling conflicts or issues with the phase;
   d. Changes to required resources based on number of personnel on course; etc.
5. The following excel document is available to use as a template for calculations. Simply “Save as” upon opening and input data as the board progresses. Once complete, insert a copy of the table for printed versions with an embedded version of the file included for use after course amendments.

Annex G - Scheduling Resources_v3.xlsx

EO 002.01	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001 Assess Ship's Stability				**Total PO time:**		
PO ATT:		**PO Admin Time:**		**PO PT time:**		
EO 001.01	Identify Stability States and forces acting on a ship	Instructor	Classroom	600		
EO 001.01	Identify Stability States and forces acting on a ship	Guest Speaker (Nav Arch)	Classroom	50		
EO 001.02	Extract information from hydrostatic curves	Instructor	Classroom	400		
EO 001.03	Calculate mean draught of a ship following a change of weight	Instructor	Classroom	800	100	
EO 001.04	Calculate the ship's center of gravity	Instructor	Classroom	1500	125	
EO 001.05	Calculate the free surface effect in a ship's compartment	Instructor	Classroom	300		
EO 001.06	Calculate the rise in G due to FCE	Instructor	Classroom	400	115	
EO 001.07	Calculate stability following an abnormal condition	Instructor	Classroom	950	115	210
PO 002 Perform the duties of Fuelling I/C				**Total PO time:**		
PO ATT:		**PO Admin Time:**		**PO PT time:**		
EO 002.01	Complete Fuelling Orders	Instructor	Classroom	100	70	

EO 002.02	Lead Maritime Vessel Fuelling	Instructor	Classroom	60		
EO 002.02	Lead Maritime Vessel Fuelling	Instructor	BCDS		500	600
PO 002: (Performance Statement)				**Total PO time:**		
PO ATT:		**PO Admin Time:**		**PO PT time:**		
EO 003.01	Act as DCPO	Instructor	Classroom	405		
EO 003.01	Act as DCPO	Instructor	DCTF		1400	
EO 003.01 EO 003.02	Act as DCPO Submit Post Incident Report	Instructor Instructor	DCTF	 300	 400	700

ANNEX H - *ADDITIONAL TRAINING RESOURCES*

In accordance with Chapter 1 para3-4, add any extra tables or paragraphs used for training resources in this annex. (**Annex H not to be deleted**)

PO 001 Assess Ship's Stability

1. Stability Extra Practice Guidebook – a book which contains extra practice for calculations trained through EO 001.01 – EO 001.07. This book will provide sufficient opportunities to trainees to work through calculations and is to be handed to trainees at the beginning of PO 001.

ANNEX I - *COSTING INFORMATION*

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

Estimates only; based on CF Cost Factors Manual FY 2012/13verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: Mar Tech PO2 MechSupr USQ		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	3	$1200		$3600	$3600	$7200
R&Q	3	$36.0	44	$4752	$4752	$47520
Incidentals	3	$17.3	44	$2284	$2284	$22840
Subtotal				**$10636**	**$10636**	**$77560**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	10	$50.0	1	$500	$500	$5000
Classroom (square meters) to include DCTF	30	$13.00	44	…	…	…
Computer requirements	11	$1,000.0	1	$11000	$0	$11000
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Instructional Materials (Printed / Handouts etc)	10	$30.0	1	$300	$300	$3000
Subtotal				**$22800**	**$800**	**$30000**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
		subtotal		**$**	**$**	**$**
	Total Estimated Costs			**$33436**	**$11436**	**$107560**

ANNEX J - *COURSE HISTORY*

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB 16 Apr – 27 Apr 18	The Marine Technician PO2 Mechanical Supervisor, follows the Marine Technician RQ PO2. Content on this course is specific to Mechanical Supervisor and is an amalgamation of training from HT and Mar Eng QL6 – QL7 Legacy training. Defining required training for this course has been particularly challenging because the RQ PO1 and RQ PO2 QSPs were designed prior to the Mechanical Supervisor course. As a result, the board had to refer back to these QSPs regularly to ensure alignment in performance was correct. In the end, full POs or portions of POs had to be moved to the PO2 Mechanical Supervisor course based on direction from D Nav P&T. Follow on work to be completed after this QSP will be to review and amend the RQ PO1 and PO2 QSPs to ensure any changes as a result of this QSPWB are actioned accordingly in those QSPs.

ANNEX K - *ASSESSMENT PLAN*

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.03	EO 001.03	To assess the trainee's ability to calculate mean draught of a ship following a change of weight.	Witten assessment	60% pass mark. Partial marks awarded for procedure in calculations	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.04	EO 001.04	To assess the trainee's ability to calculate the ship's center of gravity	Written assessment	60% pass mark. Partial marks awarded for procedure in calculations	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.06	EO 001.05, 001.06	To assess the trainee's ability to calculate the rise in G due to FSE and FCE	Written assessment	Partial marks awarded for procedure in calculations. 60% pass mark	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.07	EO 001.07	To assess the trainee's ability to calculate ship's stability due to an abnormal condition	Written assessment	Partial marks awarded for procedure in calculations. 60% pass mark	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 001	PO 001	To assess the trainee's ability to assess ship's stability	Written assessment	Partial marks awarded for procedure in calculations. 60% pass mark	First PC Fail results in a retest. Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 002.01	EO 002.01	To assess the trainee's ability to complete fueling orders	Practical	Pass/Fail 70% with feedback provided by instructor.	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.02	EO 002.02	To assess the trainee's ability to lead vessel fueling	Practical	Pass/Fail 70% IAW EC 002.02 checklist	Trainee issued Trainee Experiencing Difficulty (TED)

					and provided additional training/instruction.
PC 002	EO 002.01, 002.02	To assess the trainee's ability to perform the duties of fueling I/C	Practical	Pass/Fail 70% IAW PC 002 checklist	First PC Fail results in a retest. Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 003.01	EO 003.01	To assess the trainee's ability to act as DCPO	Practical	Pass/Fail 85% IAW PC 003/EC 003.01 checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 003.02	EO 003.02	To assess the trainee's ability to submit a post-incident report	Practical	Pass/Fail 60% IAW EC 003.02 checklist	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 003	EO 003.01	To assess the trainee's ability to Manage the DC HQ	Practical	Pass/Fail 85% IAW PC 003/EC 003.01 checklist	First PC Fail results in a retest. Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

ANNEX L - *TEL PLAN*

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
001.01, 001.04 - 001.07	**Stability Videos**: existing videos (YouTube) that demonstrate stability failures on various classes of civilian and military vessels. Updated videos can provide greater fidelity to stability factors thus increasing trainee understanding and application of concepts to calculations and recommendations. Proprietary considerations exist.	**VHS videos**: videos currently exist but are dated (B&W). Content remains unchanged but platforms and updates to today's technology. **NWO Stability Videos:** NWO uses videos to instruct stability. Developers will need to assess if these may be suitable to support instruction of these EOs	Trainees will not see modern ships affected by poor stability and damage. Poor quality of current videos impacts trainee retention as millennials expect and are used to far greater reality in animations and videos.
001.01, 001.04 - 001.07	**Interactive stability digital modelling software**: software will demonstrate the effects to stability of various loading changes to the ship (multiple platforms). The instructor or trainees should be able to manipulate factors in real time to observe impacts to stability at the ship level and further granularity to specific areas or systems.	**Physical docking models**: see-through tank with ship and block models that can be filled and emptied to demonstrate various states of stability. Ship's loads can be changed to demonstrate effects to stability. (currently available but not used given high level of instructor prep)	Trainees will not be able to properly visualize the effects of stability changes. Training without this visualization will take trainees longer to make connections between their calculations and impact to ship's stability.
002.02	**BDCS Fuelling Scenarios:** multiple fuelling scenarios covering all fuelling types which encompassing all functionality required of the BDCS to imitate on job performance to the fullest extent possible. (RAS, alongside, TAC RAS etc).	**Single Fuelling Scenario:** Both TEs currently have one scenario on BDCS that can be used to assess trainees.	Exposure to the same scenario reduces training value as each trainee is assessed. Trainees won't be exposed to all scenarios which reduces transfer of performance on the job. Assessment is difficult to complete as EC and PC will use be based on the same scenario.

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
003.01	**Emergency Stations Video**: all-inclusive footage from the initial pipe of the DC event to surveying damage following the event. Focus will be on HQ1 and SBT I/Cs. Live footage to be done in an HMC ship with animations to fill any gaps. Sea training to be used for SME input.	**Experience**: unlikely that the trainees have seen the HQ1 position before. Gaps will be trained by activating their knowledge as SBT I/C to scaffold their knowledge and performance as DCPO.	Trainees will not have a standardized "perfect demonstration" of how to be DCPO which can lead to incorrect performance of the expected standard.